



7 Series FPGAs and Zynq-7000 SoC XADC Dual 12-Bit 1 MSPS Analog-to-Digital Converter User Guide (UG480)

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About This Guide

Guide Contents

Additional Resources

Introduction and Quick Start

- XADC Overview

- XADC Pinout Requirements

- Instantiating the XADC

Analog-to-Digital Converter

- ADC Transfer Functions

- Analog Inputs

XADC Register Interface

- Status Registers

- Control Registers

- DRP JTAG Interface

- Zynq-7000 SoC Processing System (PS) to XADC Dedicated Interface

XADC Operating Modes

- Single Channel Mode

- Automatic Channel Sequencer

- Sequencer Modes

- External Multiplexer Mode

- Maximum and Minimum Status Registers

- Automatic Alarms

XADC Timing

- Continuous Sampling

- Event-Driven Sampling

- Dynamic Reconfiguration Port (DRP) Timing

Application Guidelines

- Reference Inputs (VREFP and VREFN)

- Analog Power Supply and Ground (VCCADC and GNDADC)

- External Analog Inputs

- PC Board Design Guidelines

Additional Resources and Legal Notices

- Xilinx Resources

- Solution Centers

- Documentation Navigator and Design Hubs

- References

- Revision History

- Please Read: Important Legal Notices

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About This Guide

Xilinx® 7 series FPGAs include four FPGA families that are all designed for lowest power to enable a common design to scale across families for optimal power, performance, and cost. The Spartan®-7 family is the lowest density with the lowest cost entry point into the 7 series portfolio. The Artix®-7 family is optimized for highest performance-per-watt and bandwidth-per-watt for cost-sensitive, high volume applications. The Kintex®-7 family is an innovative class of FPGAs optimized for the best price-performance. The Virtex®-7 family is optimized for highest system performance and capacity. The Zynq®-7000 SoC device integrates a feature-rich dual-core Arm® Cortex™-A9 based processing system (PS) and 28 nm Xilinx programmable logic (PL) in a single device.

This guide serves as a technical reference describing the 7 series FPGAs and Zynq-7000 SoC XADC, a dual 12-bit, 1 MSPS analog-to-digital converter with on-chip sensors. This user guide is part of an overall set of documentation on the 7 series FPGAs and Zynq-7000 SoC devices, which is available on the Xilinx website at www.xilinx.com/documentation.

Guide Contents

This manual contains these chapters:

- [Chapter 1, Introduction and Quick Start](#)
- [Chapter 2, Analog-to-Digital Converter](#)
- [Chapter 3, XADC Register Interface](#)
- [Chapter 4, XADC Operating Modes](#)
- [Chapter 5, XADC Timing](#)
- [Chapter 6, Application Guidelines](#)
- [Appendix 7, Additional Resources and Legal Notices](#)

Additional Resources

To find additional Xilinx documentation, see the Xilinx website at:

www.xilinx.com/support/documentation/index

To search the Answer Database of silicon, software, and IP questions and answers, see the Xilinx website at:

www.xilinx.com/support

To find Xilinx Analog Mixed Signal (AMS) information, see the Xilinx website at:

www.xilinx.com/ams

Example design files and Tcl console examples for the Vivado® Hardware Manager can be found in the [ZIP file](#) that accompanies this user guide.

Introduction and Quick Start

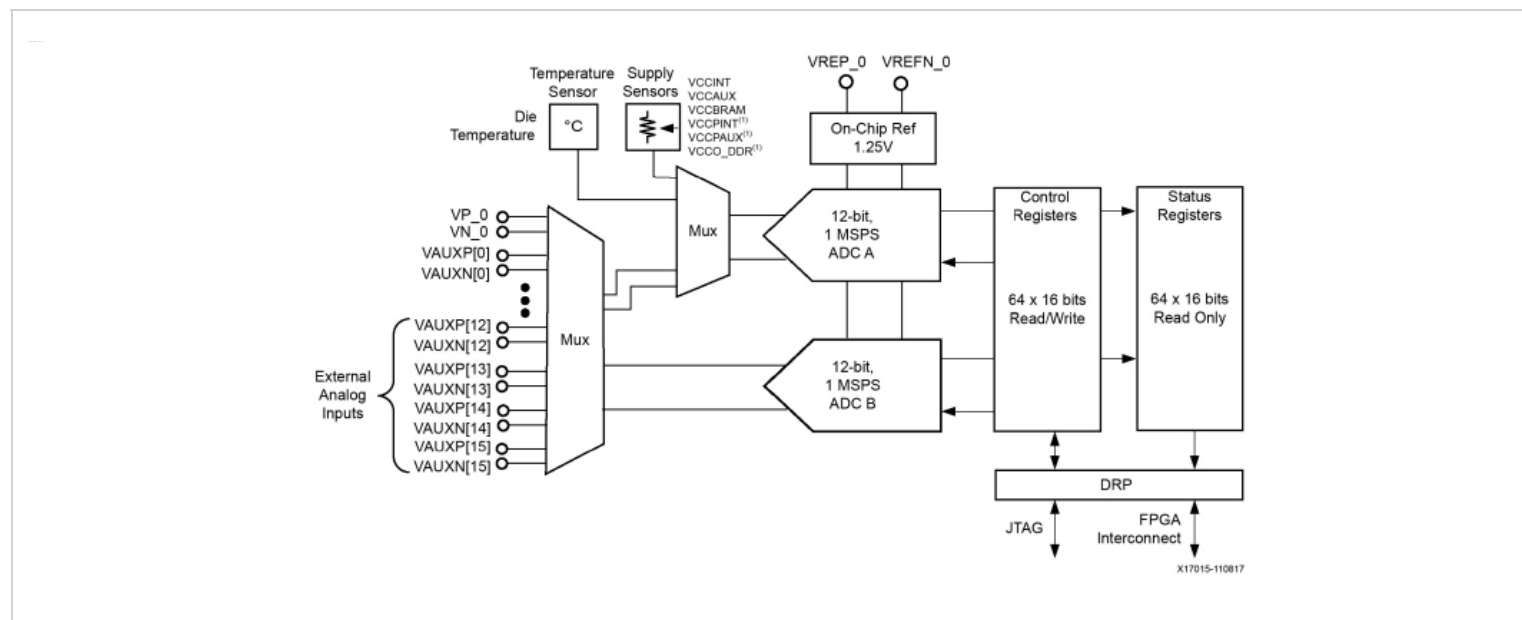
This chapter provides a brief overview of the Xilinx 7 series FPGAs XADC functionality. The XADC is available in all Artix®-7, Kintex®-7, Virtex®-7, and Zynq®-7000 SoC devices. The XADC is also available in many, but not all Spartan®-7 devices. To identify specific devices that support the XADC block, consult the Spartan-7 family overview in DS180, *7 Series FPGAs Overview* [Ref 11] .

The XADC is the basic building block that enables analog mixed signal (AMS) functionality which is new to 7 series FPGAs. By combining high quality analog blocks with the flexibility of programmable logic, it is possible to craft customized analog interfaces for a wide range of applications. See www.xilinx.com/ams for more information.

This chapter contains only key information to allow a basic understanding of the XADC block. With this introduction, you can learn the pinout requirements and determine how to instantiate basic functionality in their designs. Subsequent chapters provide more detailed descriptions of the XADC functionality.

XADC Overview

The XADC includes a dual 12-bit, 1 Mega sample per second (MSPS) ADC and on-chip sensors. The ADCs and sensors are fully tested and specified (see the respective 7 series FPGAs data sheet). The ADCs provide a general-purpose, high-precision analog interface for a range of applications. [Figure 1-1](#) shows a block diagram of the XADC. The dual ADCs support a range of operating modes, for example, externally triggered and simultaneous sampling on both ADCs (see [Chapter 4, XADC Operating Modes](#)) and various analog input signal types, for example, unipolar and differential (see [Chapter 2, Analog-to-Digital Converter](#)). The ADCs can access up to 17 external analog input channels.



XADC Block Diagram

Notes relevant to [Figure 1-1](#) :

1. Zynq-7000 SoC devices only.

The XADC also includes several on-chip sensors that support measurement of the on-chip power supply voltages and die temperature. The ADC conversion data is stored in dedicated registers called status registers. These registers are accessible through the FPGA interconnect using a 16-bit synchronous read and write port called the dynamic reconfiguration port (DRP). ADC conversion data is also accessible through the JTAG TAP, either before (pre-configuration) or after configuration. For JTAG TAP, users are not required to instantiate the XADC because it is a dedicated interface that uses the existing FPGA JTAG infrastructure. As discussed later, if the XADC is not instantiated in a design, the device operates in a predefined mode (called

default mode) that monitors on-chip temperature and supply voltages.

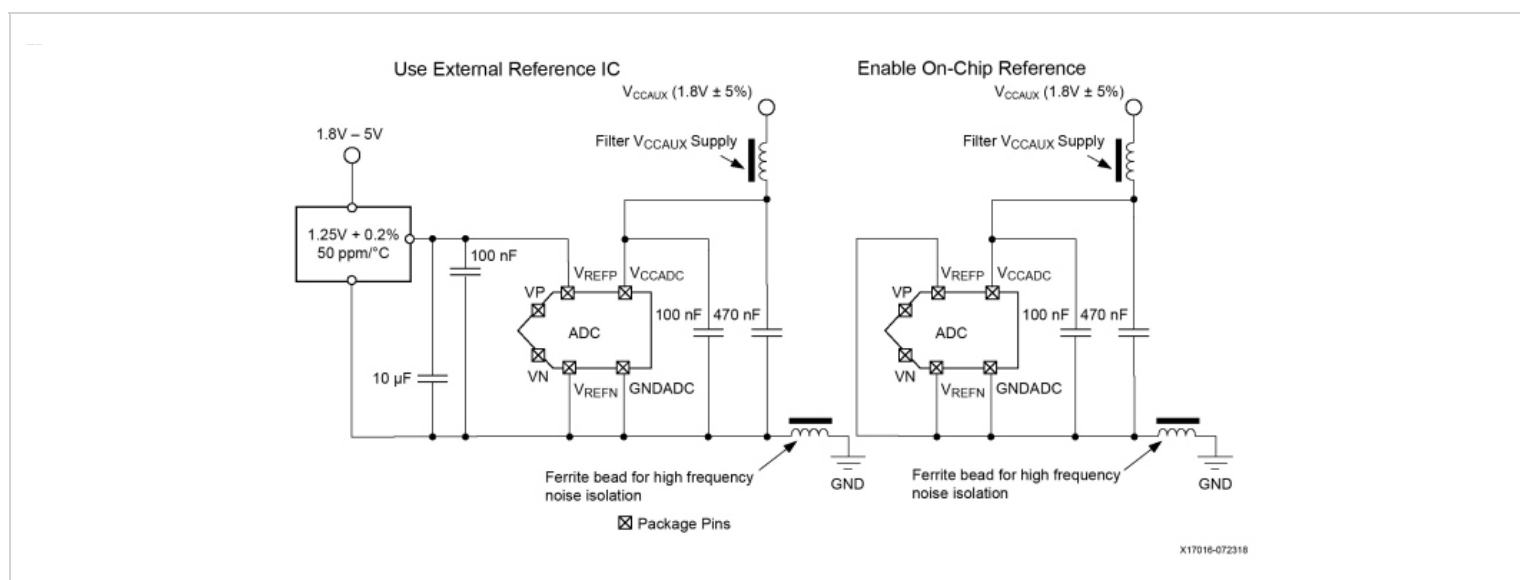
XADC operation is user defined by writing to the control registers using either the DRP or JTAG interface. It is also possible to initialize these register contents when the XADC is instantiated in a design using the block attributes.

For Virtex-5 and Virtex-6 FPGA System Monitor users, the XADC functionality is fully backward compatible with legacy System Monitor designs. The XADC functionality and interface are familiar to those who have previously designed with the System Monitor. System Monitor designs are automatically retargeted to the XADC site by the software tools.

However, the XADC block in 7 series FPGAs contains a large number of new features and enhancements detailed in subsequent chapters. The new functionality is enabled by initializing previously undefined status registers and bit locations. Old System Monitor designs that did not initialize these new registers or bit locations behave exactly the same way as before.

XADC Pinout Requirements

All XADC dedicated pins are located in bank 0 and thus have the `_0` suffix in the package file names. [Figure 1-2](#) shows the basic pinout requirements for the XADC. There are two recommended configurations. On the left, the XADC is powered from V_{CCAUX} (1.8V) and uses an external 1.25V reference source. The external reference delivers the best performance in terms of accuracy and thermal drift. A ferrite bead is used to isolate the ground reference for the analog circuits and system ground. An additional low-pass filter for V_{CCAUX} supply will similarly improve the ADC performance. See Chapter 6, Application Guidelines for more information. Shared or common ground impedance is the most common way to introduce unwanted noise into analog circuits.



XADC Pinout Requirements

It is also possible to use an on-chip reference for the ADCs. To enable the on-chip reference source, the V_{REFP} pin must be connected to ground as shown on the right of [Figure 1-2](#). Where only basic on-chip thermal and supply monitoring is required, using the on-chip reference provides good performance. Users should consult the respective data sheet to see the accuracy specifications when using the external and on-chip reference sources. [Table 1-1](#) lists the pins associated with the XADC and the recommended connectivity.

Note: It is also important to place the 100 nF decoupling capacitors as close as possible to the package balls to minimize inductance between the decoupling and package balls.

Table: XADC Package Pins

Package Pin	Type	Description
V_{CCADC_0}	Power supply	This is the analog supply pin for the ADCs and other analog circuits in the

		XADC. It can be tied to the 1.8V V_{CCAUX} supply; however, in a mixed-signal system, the supply should be connected to a separate 1.8V analog, if available. See Analog Power Supply and Ground (V_{CCADC} and GNDADC), page 65 for more information. This pin should never be tied to GND. The pin should be tied to V_{CCAUX} even if the XADC is not being used.
GNDADC_0	Power supply	This is the ground reference pin for the ADCs and other analog circuits in the XADC. It can be tied to the system ground through an isolating ferrite bead as shown in Figure 1-2. In a mixed-signal system this pin should be tied to an analog ground plane if available, in which case the ferrite bead is not required. See Analog Power Supply and Ground (V_{CCADC} and GNDADC), page 65 for more information. This pin should always be tied to GND even if the XADC is not being used.
V_{REFP_0}	Reference voltage input	This pin can be tied to an external 1.25V accurate reference IC ($\pm 0.2\%$ or ± 9 LSBs at 12 bits) for best performance of the ADCs. It should be treated as an analog signal that together with the V_{REFN} signal provides a differential 1.25V voltage. By connecting this pin to GNDADC (see Figure 1-2), an on-chip reference source ($\pm 1\%$ or ± 41 LSBs at 12 bits) is activated. This pin should always be connected to GNDADC if an external reference is not supplied. See Reference Inputs (V_{REFP} and V_{REFN}), page 65 for more information.
V_{REFN_0}	Reference voltage input	This pin should be tied to the GND pin of an external 1.25V accurate reference IC ($\pm 0.2\%$) for best performance of the ADCs. It should be treated as an analog signal that together with the V_{REFP} signal provides a differential 1.25V voltage. This pin should always be connected to GND even if an external reference is not supplied. See Reference Inputs (V_{REFP} and V_{REFN}), page 65 for more information.

V_{P_0}	Dedicated analog input	This is the positive input terminal of the dedicated differential analog input channel (V_P/V_N). The analog input channels are very flexible and support multiple analog input signal types. For more information, see Analog Inputs, page 21 . This pin should be connected to GND if not used.
V_{N_0}	Dedicated analog input	This is the negative input terminal of the dedicated differential analog input channel (V_P/V_N). The analog input channels are very flexible and support multiple analog input signal types. For more information, see Analog Inputs, page 21 . This pin should be connected to GND if not used.
$_AD0P_$ to $_AD15P_$ (1) (2)	Auxiliary analog inputs/digital I/O	These are multi-function pins that can support analog inputs or can be used as regular digital I/O (see Figure 1-1). These pins support up to 16 positive input terminals of the differential auxiliary analog input channels (V_{AUXP}/V_{AUXN}). The analog input channels are very flexible and support multiple analog input signal types. For more information, see Analog Inputs, page 21 . When not being used as analog inputs, these pins can be treated like any other digital I/O.
$_AD0N_$ to $_AD15N_$ (1) (2)	Auxiliary analog inputs/digital I/O	These are multi-function pins that can support analog inputs or can be used as regular digital I/O (see Figure 1-1). These pins support up to 16 negative input terminals of the differential auxiliary analog input channels (V_{AUXP}/V_{AUXN}). The analog input channels are very flexible and support multiple analog input signal types. For more information, see Analog Inputs, page 21 . When not being used as analog inputs, these pins can be treated like any other digital I/O.

Notes:

1. FPGA I/Os that are analog input-enabled contain the $_ADxP_$ and $_ADxN_$ designation in the package file name, for example, $IO_L1P_T0_AD0P_35$ is the input pin for analog auxiliary channel $VAUXP[0]$. $IO_L1N_T0_AD0N_35$ is the input pin for analog auxiliary channel $VAUXN[0]$. For more information, see UG475, *7 Series FPGAs Packaging and Pinout Product Specifications User Guide* [\[Ref 2\]](#).

2. Auxiliary channels 6, 7, 13, 14, and 15 are not supported in Kintex-7 devices. Some auxiliary analog channels might also not be supported in certain Virtex-7, Artix-7, Spartan-7, and Zynq-7000 SoC device package options. You should consult the package file for the device.

Note: Chapter 6, Application Guidelines, should be consulted before commencing any PC board layout. Board layout and external component choices can greatly impact the performance of the ADCs.

Apart from a single dedicated analog input pair (V_P/V_N), the external analog inputs use dual-purpose I/O. These FPGA digital I/Os are individually nominated as analog inputs when the XADC is instantiated in a design. This document refers to these analog inputs as auxiliary analog inputs. A maximum of 16 auxiliary analog inputs are available. The auxiliary analog inputs are enabled by connecting the analog inputs on the XADC primitive to the top level of the design. When enabled as analog inputs, these package balls are unavailable as digital I/Os. It is also possible to enable the auxiliary analog inputs preconfiguration (for example, for PCB diagnostics) through the JTAG TAP (see [JTAG DRP Commands](#) for more information.)

All analog input channels are differential and require two package balls. Typically, the auxiliary analog inputs are allocated evenly over banks 15 and 35. However, users should consult the pinout information in UG475, *7 Series FPGAs Packaging and Pinout Product Specifications User Guide* [Ref 2] for a particular device and package combination. Analog-capable I/O have the $ADxP$ or $ADxN$ suffix on the I/O name in the package files. For example, auxiliary analog input channel 8 has associated package ball names ending with AD8P and AD8N. See UG475, *7 Series FPGAs Packaging and Pinout Product Specifications User Guide* [Ref 2] for more information. The auxiliary analog inputs have a fixed package ball assignment and cannot be moved.

Auxiliary analog inputs are supported differently in Vivado® tools when compared to ISE tools. The auxiliary analog inputs do not require any user-specified constraints or pin locations in ISE tools. ISE external auxiliary inputs do not need an I/O standard setting to be added to your constraints file (UCF) or in the PlanAhead™ design tool. In Vivado design tools, the auxiliary analog inputs must be assigned to the associated pin location.

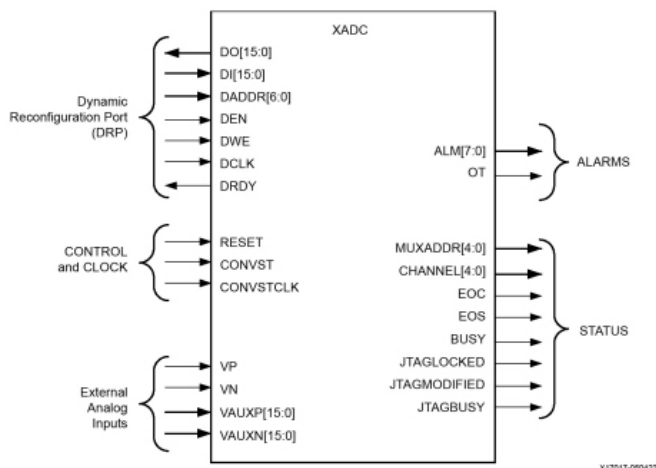
Auxiliary analog inputs must be connected to the top level of the design.

Note: Auxiliary channels 6, 7, 13, 14, and 15 are not supported on Kintex-7 devices. Some auxiliary analog channels might also not be supported in certain Virtex-7, Artix-7, Spartan-7, and Zynq-7000 SoC device package options. Users should consult the package file for the device.

Instantiating the XADC

As mentioned previously, it is not necessary to instantiate the XADC in a design to access the on-chip monitoring capability. However, if the XADC is not instantiated in a design, the only way to access this information is through the JTAG test access port (TAP). To allow access to the status registers (measurement results) from the FPGA logic, the XADC must be instantiated. These subsections give a brief overview of the XADC primitive (ports and attributes).

[Figure 1-3](#) shows the ports on the XADC primitive, and [Table 1-2](#) describes the functionality of the ports.



XADC Primitive Ports

Table: XADC Port Descriptions

Port	I/O	Description
DI[15:0]	Inputs	Input data bus for the DRP. ⁽¹⁾
DO[15:0]	Outputs	Output data bus for the DRP. ⁽¹⁾
DADDR[6:0]	Input	Address bus for the DRP. ⁽¹⁾
DEN ⁽²⁾	Input	Enable signal for the DRP. ⁽¹⁾
DWE ⁽²⁾	Input	Write enable for the DRP. ⁽¹⁾
DCLK	Input	Clock input for the DRP. ⁽¹⁾
DRDY ⁽²⁾	Output	Data ready signal for the DRP. ⁽¹⁾
RESET ⁽²⁾	Input	Asynchronous reset signal for the XADC control logic. RESET will be deasserted synchronously to DCLK or the internal configuration clock when DCLK is stopped.
CONVST ⁽³⁾	Input	Convert start input. This input controls the sampling instant on the ADC(s) inputs and is only used in event mode timing (see Event-Driven Sampling, page 62). This input comes from the general-purpose interconnect in the FPGA logic.
CONVSTCLK ⁽³⁾	Input	Convert start clock input. This input is connected to a clock net. Like CONVST, this input controls the sampling instant on the ADC(s) inputs and is only used in event mode timing. This input comes from the local clock distribution network in the FPGA logic. Thus, for the best control over the

		sampling instant (delay and jitter), a global clock input can be used as the CONVST source.
V_P, V_N	Input	One dedicated analog input pair. The XADC has one pair of dedicated analog input pins that provide a differential analog input. When designing with the XADC feature but not using the dedicated external channel of V_P and V_N , you should connect both V_P and V_N to analog ground.
VAUXP[15:0], VAUXN[15:0]	Inputs	Sixteen auxiliary analog input pairs. In addition to the dedicated differential analog input, the XADC can access 16 differential analog inputs by configuring digital I/O as analog inputs. These inputs can also be enabled pre-configuration through the JTAG port (see DRP JTAG Interface, page 40).
ALM[0] ⁽²⁾	Output	Temperature sensor alarm output.
ALM[1] ⁽²⁾	Output	V_{CCINT} sensor alarm output.
ALM[2] ⁽²⁾	Output	V_{CCAUX} sensor alarm output.
ALM[3] ⁽²⁾	Output	V_{CCBRAM} sensor alarm output.
ALM[4] ⁽⁴⁾	Output	V_{CCPINT} sensor alarm output.
ALM[5] ⁽⁴⁾	Output	V_{CCPAUX} sensor alarm output.
ALM[6] ⁽⁴⁾	Output	V_{CCO_DDR} sensor alarm output.
ALM[7] ⁽²⁾	Output	Logic OR of bus ALM[6:0]. Can be used to flag the occurrence of any alarm.
OT ⁽²⁾	Output	Over-Temperature alarm output.
MUXADDR[4:0]	Outputs	These outputs are used in external multiplexer mode. They indicate the address of the next channel in a sequence to be converted. They provide the channel address for an external multiplexer (see External Multiplexer Mode, page 53).
CHANNEL[4:0]	Outputs	Channel selection outputs. The ADC input MUX channel selection for the current ADC conversion is placed on these outputs at the end of an ADC conversion.

EOC ⁽²⁾	Output	End of conversion signal. This signal transitions to active-High at the end of an ADC conversion when the measurement is written to the status registers (see Chapter 5, XADC Timing).
EOS ⁽²⁾	Output	End of sequence. This signal transitions to active-High when the measurement data from the last channel in an automatic channel sequence is written to the status registers (see Chapter 5, XADC Timing).
BUSY ⁽²⁾	Output	ADC busy signal. This signal transitions High during an ADC conversion. This signal also transitions High for an extended period during an ADC or sensor calibration.
JTAGLOCKED ⁽²⁾	Output	Indicates that a DRP port lock request has been made by the JTAG interface (see DRP JTAG Interface, page 40). This signal is also used to indicate that the DRP is ready for access (when Low).
JTAGMODIFIED ⁽²⁾	Output	Used to indicate that a JTAG write to the DRP has occurred.
JTAGBUSY ⁽²⁾	Output	Used to indicate that a JTAG DRP transaction is in progress.

Notes:

1. The DRP is the interface between the XADC and FPGA. All XADC registers can be accessed from the FPGA logic using this interface. For more details on the timing for these DRP signals, see [Figure 5-3, page 64](#).
2. Active-High signal.
3. Rising edge triggered signal.
4. Only available on Zynq-7000 SoC devices.

The block diagram in [Figure 1-1](#) shows the control registers that define the operation of the XADC. The control registers are a set of 32 16-bit registers. As mentioned, these registers can be read and written through the DRP or JTAG ports. It is also possible to initialize the contents of these registers during the configuration of the FPGA. This enables the XADC to start operating in a user-defined mode after configuration is complete. There are 32 attributes associated with the XADC primitive that allow users to initialize these registers. [Table 1-3](#) lists these attributes. The attributes are called INIT_xx, where xx corresponds to the hexadecimal address of the register on the DRP. For example, INIT_40 corresponds to the first control register at address 40h on the DRP.

Table: XADC Primitive Attributes

Attribute	Name	Control Register Address	Description
INIT_40	Configuration register 0	40h	XADC configuration registers (see Control Registers ,

INIT_41	Configuration register 1	41h	page 35).
INIT_42	Configuration register 2	42h	
INIT_43 to INIT_47	Test registers	43h to 47h	XADC Test registers for factory use only. The default initialization is 0000h .
INIT_48 to INIT_4F	Sequence registers	48h to 4Fh	Sequence registers used to program the XADC Channel Sequencer function (see Automatic Channel Sequencer, page 46).
INIT_50 to INIT_5F	Alarm limit registers	50h to 5Fh	Alarm threshold registers for the XADC alarm function (see Automatic Alarms, page 56).

The XADC primitive also has an attribute called SIM_MONITOR_FILE that points to the analog stimulus file. This attribute is required to support simulation. This attribute points to the path and file name of a text file that contains analog information (for example, temperature and voltage). UNISIM and SIMPRIM models use this text file during simulation. This is the only way analog signals can be introduced into a simulation of the XADC. For more information see [XADC Software Support, page 70](#) .

Instantiating the XADC involves connecting the required I/O (including analog inputs) to the design and optionally initializing the control registers to define the XADC operation after configuration. Alternatively, users can write to the control registers through the DRP after device configuration. The timing diagram for DRP read and write operations is shown in [Figure 5-3](#) .

Note: The read/write operation is not valid or complete until the DRDY signal goes active.

This subsection provides a brief example of an XADC instantiation using Verilog. First, the control registers are initialized, and then the required XADC I/Os are connected to the design. The software correctly ties off unconnected I/Os on the primitive.

This design assumes an external 50 MHz clock is used for DCLK, and the XADC is configured to monitor temperature, supply voltages, and activate alarms if safe limits are exceeded. This example is explained in detail in [XADC Software Support, page 70](#)

```

XADC #(
    // Initializing the XADC Control Registers
    .INIT_40(16'h9000), // Calibration coefficient averaging disabled
    // averaging of 16 selected for external channels
    .INIT_41(16'h2ef0), // Continuous Sequencer Mode, Disable unused ALMs,
    // Enable calibration
    .INIT_42(16'h0400), // Set DCLK divider to 4, ADC = 500Ksps, DCLK = 50MHz
    .INIT_48(16'h4701), // Sequencer channel - enable Temp sensor, VCCINT, VCCAUX,
    // VCCBRAM, and calibration
    .INIT_49(16'h000f), // Sequencer channel - enable aux analog channels 0 - 3
    .INIT_4A(16'h4700), // Averaging enabled for Temp sensor, VCCINT, VCCAUX,
    // VCCBRAM
    .INIT_4B(16'h0000), // No averaging on external channels
    .INIT_4C(16'h0000), // Sequencer Bipolar selection
    .INIT_4D(16'h0000), // Sequencer Bipolar selection
    .INIT_4E(16'h0000), // Sequencer Acq time selection
    .INIT_4F(16'h0000), // Sequencer Acq time selection
    .INIT_50(16'hb5ed), // Temp upper alarm trigger 85°C
    .INIT_51(16'h5999), // Vccint upper alarm limit 1.05V
    .INIT_52(16'ha147), // Vccaux upper alarm limit 1.89V
    .INIT_53(16'h0000), // OT upper alarm limit 125°C using automatic shutdown
    .INIT_54(16'ha93a), // Temp lower alarm reset 60°C
    .INIT_55(16'h5111), // Vccint lower alarm limit 0.95V
    .INIT_56(16'h91Eb), // Vccaux lower alarm limit 1.71V
    .INIT_57(16'hae4e), // OT lower alarm reset 70°C
    .INIT_58(16'h5999), // VCCBRAM upper alarm limit 1.05V
    .INIT_5C(16'h5111), // VCCBRAM lower alarm limit 0.95V
    .SIM_MONITOR_FILE("sensor_input.txt")
    // Analog Stimulus file. Analog input values for simulation
)
XADC_INST ( // Connect up instance IO. See UG480 for port descriptions

```

```

.CONVST(GND_BIT),      // not used
.CONVSTCLK(GND_BIT),   // not used
.DADDR(DADDR_IN[6:0]),
.DCLK(DCLK_IN),
.DEN(DEN_IN),
.DI(DI_IN[15:0]),
.DWE(DWE_IN),
.RESET(RESET_IN),
.VAUXN(aux_channel_n[15:0]),
.VAUXP(aux_channel_p[15:0]),
.ALM(alm_int),
.BUSY(BUSY_OUT),
.CHANNEL(CHANNEL_OUT[4:0]),
.DO(DO_OUT[15:0]),
.DRDY(DRDY_OUT),
.EOC(EOC_OUT),
.EOS(EOS_OUT),
.JTAGBUSY(),           // not used
.JTAGLOCKED(),         // not used
.JTAGMODIFIED(),       // not used
.OT(OT_OUT),
.MUXADDR(),            // not used
.VP(VP_IN),
.VN(VN_IN)
);

```

More comprehensive information regarding the operation of the ADCs and on-chip sensors can be found in [Chapter 2, Analog-to-Digital Converter](#) . This section provides a brief overview to help users to quickly interpret data read from the status registers and verify the operation of the XADC.

The ADCs have a nominal analog input range from 0V to 1V. In unipolar mode (default), the analog inputs of the ADCs produce a full scale code of FFFh (12 bits) when the input is 1V. Thus, an analog input signal of 200 mV in unipolar mode produces and outputs code of [Equation 1-1](#) .

$$((0.2 / 1.0) \times \text{FFFh}) = 819 \text{ or } 333\text{h}$$

In bipolar mode, the ADCs use two's complement coding and produces a full scale code of 7FFh with +0.5V input and 800h with -0.5V input.

The temperature sensor has a transfer function given by [Equation 1-2](#) .

$$\text{Temp } (^{\circ}\text{C}) = \frac{\text{ADC Code} \times 503.975}{4096} - 273.15$$

For example, ADC Code 2423 (977h) = 25°C.

The temperature sensor result can be found in status register 00h .

The XADC power supply sensors have a transfer function that generates a full scale ADC output code of FFFh with a 3V input voltage. This voltage is outside the allowed supply range, but the FPGA supply measurements map into this measurement range. Thus, $V_{\text{CCINT}} = 1\text{V}$ generates an output code of $1/3 \times 4096 = 1365 = 555\text{h}$. The XADC monitors V_{CCINT} , V_{CCAUX} , and V_{CCBRAM} . The measurement results are stored in status registers 01h , 02h , and 06h , respectively.

The XADC monitors three additional power supplies on the Zynq-7000 SoC devices. The supplies are V_{CCPINT} , V_{CCPAUX} , and $V_{\text{CCO_DDR}}$. These measurements are stored in status registers 0Dh , 0Eh , and 0Fh , respectively.

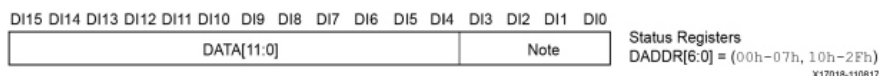
Analog-to-Digital Converter

The XADC block contains two 12-bit, 1 MSPS ADCs. These ADCs are available for use with both external analog inputs and on-chip sensors. Several predefined operating modes are available that cover the most typical use cases for these ADCs. The various operating modes are covered in [Chapter 4, XADC Operating Modes](#) . This chapter focuses on the detailed operation of the ADC and the on-chip sensors. The various input configurations for the external analog inputs are also covered. All operating modes of the ADC, sensors, and analog inputs are configured using the XADC control registers. A detailed description of the control registers is covered in [Chapter 3, XADC Register Interface](#) .

ADC Transfer Functions

The ADCs have transfer functions as shown in [Figure 2-2](#) and [Figure 2-3](#) . These transfer functions reflect unipolar and bipolar operating modes, respectively. All on-chip sensors use the unipolar mode of operation for the ADC. Users can optionally configure the external analog input channels to operate in unipolar or bipolar modes (see [Analog Inputs, page 21](#)).

For the ADCs to function as specified, the power supplies and reference options must be configured correctly. The required package ball connections are shown in [Figure 1-2](#) . Other important aspects to ensure optimal ADC performance are the PCB layout and external component selection. These issues are covered in Chapter 6, Application Guidelines . It is recommended that you read this chapter is before the board design is started.

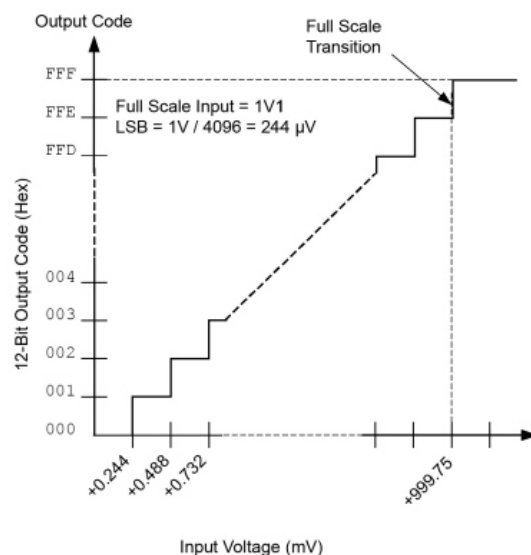


Status Registers

Note: The ADCs always produce a 16-bit conversion result. The 12-bit data correspond to the 12 MSBs (most significant) in the 16-bit status registers. The unreferenced LSBs can be used to minimize quantization effects or improve resolution through averaging or filtering. See [Figure 2-1](#) .

[Figure 2-2](#) shows the 12-bit unipolar transfer function for the ADCs. The nominal analog input range to the ADCs is 0V to 1V in this mode. The ADC produces a zero code (000h) when 0V is present on the ADC input and a full scale code of all 1s (FFFh) when 1V is present on the input.

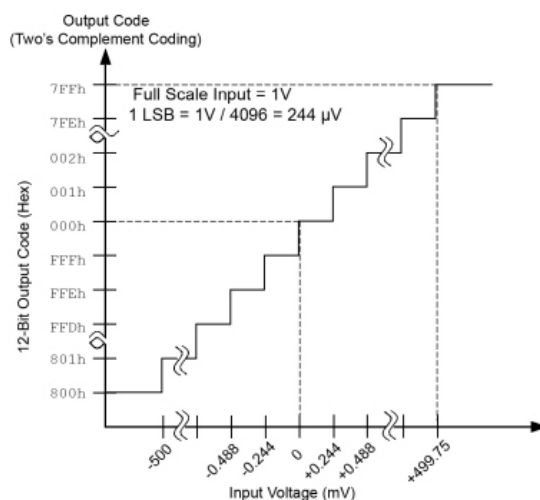
The ADC output coding in unipolar mode is straight binary. The designed code transitions occur at successive integer LSB values such as one LSB, two LSBs, and three LSBs, etc. The LSB size in volts is equal to $1V/2^{12}$ or $1V/4096 = 244 \mu V$. The analog input channels are differential in nature and require both the positive (V_P) and negative (V_N) inputs of the differential input to be driven. More details on the analog inputs and the kinds of inputs signals that can be accommodated are covered in the [Analog Inputs](#) section.



Unipolar Transfer Function

When the external analog input channels of the ADCs are configured as bipolar, they can accommodate true differential and bipolar analog signal types (see the [Analog Inputs](#) section). When dealing with differential signal types, it is useful to have both sign and magnitude information about the analog input signal. [Figure 2-3](#) shows the ideal transfer function for bipolar mode

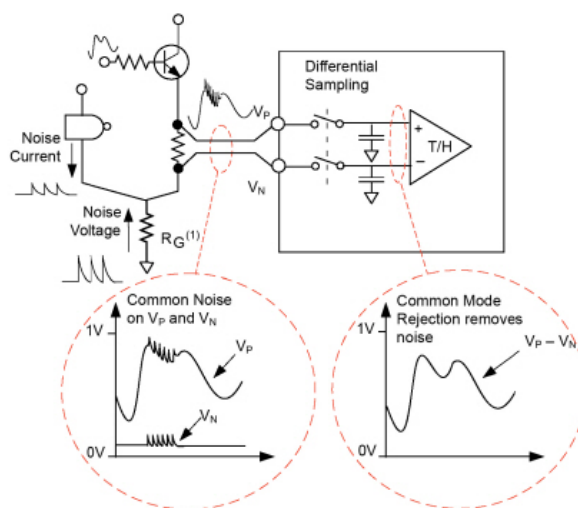
operation. The output coding of the ADC in bipolar mode is two's complement and is intended to indicate the sign of the input signal on V_P relative to V_N . The designed code transitions occur at successive integer LSB values, that is, one LSB, two LSBs, three LSBs, etc. The LSB size in volts is equal to $1V/2^{12}$ or $1V/4096 = 244 \mu V$.



Bipolar Transfer Function

Analog Inputs

The analog inputs of the ADC use a differential sampling scheme to reduce the effects of common-mode noise signals. This common-mode rejection improves the ADC performance in noisy digital environments. Figure 2-4 shows the benefits of a differential sampling scheme. Common ground impedances (R_G) couple noise voltages (switching digital currents) into other parts of a system. These noise signals can be 100 mV or more. For the ADCs, this noise voltage is equivalent to hundreds of LSBs, thus inducing large measurement errors. The differential sampling scheme samples both the signal and any common mode noise voltages at both analog inputs (V_P and V_N). The common mode signal is effectively subtracted because the Track-and-Hold amplifier captures the difference between V_P and V_N or V_P minus V_N . To take advantage of the high common mode rejection, users need only connect V_P and V_N in a differential configuration.



Note 1: R_G is Common Ground Impedance.

Common Mode Noise Rejection

The auxiliary analog inputs (VAUXP[15:0] and VAUXN[15:0]) are analog inputs that are shared with regular digital I/O package balls. The auxiliary analog inputs are automatically enabled when the XADC is instantiated in a design, and these inputs are connected on the top level of the design. The auxiliary analog inputs do not require any user-specified constraints or pin locations. They do not require an I/O standard setting to be added to the UCF or in the PlanAhead pinout tool. In the Vivado design tools, an IOSTANDARD must be selected that is compatible for the bank even though the IOSTANDARD does not affect the input programming. All configuration is automatic when the analog inputs are connected to the top level of the design. Only those auxiliary inputs connected in a design are enabled as analog inputs. The XADC auxiliary inputs pins are labeled in UG475, *7 Series FPGAs Packaging and Pinout Product Specifications User Guide* [Ref 2] by appending `_ADxP_` and `_ADxN_` to the I/O name, where *x* is the auxiliary pair number. For example, the auxiliary input VAUXP[15] could be designated `IO_LxxP_xx_AD15P_xx` in the pinout specification.

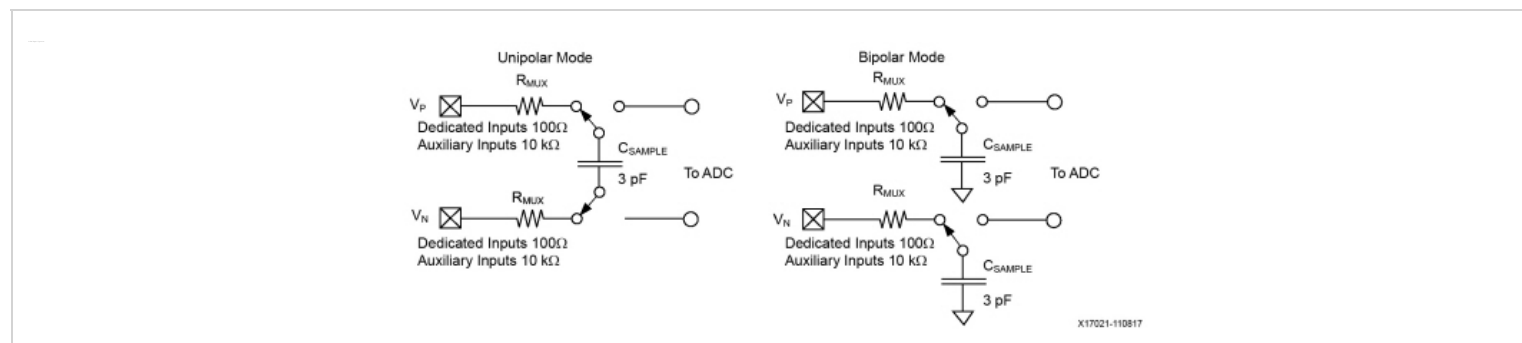
When designated as analog inputs, these inputs are unavailable for use as digital I/Os. If the I/O is used as a digital I/O, it is subject to the specifications of the configured I/O standard. If the I/O is used as an analog input, the input voltage must adhere to the specifications given in the “Analog-to-Digital Converter” section of the respective data sheet.

It is possible to enable any number of auxiliary analog inputs in an I/O bank and use the remaining as digital I/Os. If there is a mixture of analog and digital I/Os in a bank, the I/O bank must be powered by a supply required to meet the specifications of the digital I/O standard in used. The analog input signal should not exceed the I/O bank supply voltage (V_{CCO}) in this case.

Note: Auxiliary channels 6, 7, 13, 14, and 15 are not supported on Kintex-7 devices. Some auxiliary analog channels might also not be supported in certain Virtex-7, Artix-7, Spartan-7, and Zynq-7000 SoC device package options. Users should consult the package file for the device.

The maximum conversion rate specified for the ADC is 1 MSPS or a conversion time of 1 μ s. In continuous sampling mode (see [Chapter 5, XADC Timing](#)), 26 ADCCLK cycles are required to acquire an analog signal and perform a conversion. This implies a maximum ADCCLK frequency of 26 MHz. If the ACQ bit has not been set, four ADCCLKs or 150 ns is allowed for the final stages of the acquisition. When using single channel mode, the ACQ bit in configuration register 0[40h] (see [Control Registers, page 35](#)), or when using the sequencer, the appropriate ACQ bit in the sequencer registers should be set (see [ADC Channel Settling Time \(4Eh and 4Fh\), page 49](#)). This “settling” time ensures that the analog input voltage is acquired to a 12-bit accuracy. The settling time can be increased by reducing the ADCCLK frequency or setting the ACQ bit. In the latter case, assuming the 26 MHz maximum clock frequency, the settling time is increased to 380 ns (10 ADCCLK cycles), and the conversion rate would be reduced to 812 kSPS for the same ADCCLK frequency. In event timing mode (see [Chapter 5, XADC Timing](#)), you can initiate the conversion cycle by using CONVST or CONVSTCLK, allowing more control over the acquisition time, if required.

[Figure 2-5](#) illustrates an equivalent analog input circuit for the external analog input channels in both unipolar and bipolar configurations. The analog inputs consist of a sampling switch and a sampling capacitor used to acquire the analog input signal for conversion. During the ADC acquisition phase, the sampling switch is closed, and the sampling capacitor is charged up to the voltage on the analog input. The time needed to charge this capacitor to its final value (± 0.5 LSBs at 12 bits) is determined by the capacitance of the sampling capacitor (C_{SAMPLE}), the resistance of the analog multiplexer circuit (R_{MUX}), and any external (source) impedance.



Equivalent Analog Input Circuits

The required 12-bit acquisition time (assuming no additional external or source resistance) in bipolar mode for example is approximated by [Equation 2-1](#).

$$t_{ACQ} = 9 \times R_{MUX} \times C_{SAMPLE}$$

The time constant 9 is derived from $TC = \ln 2(N + m)$, where $N = 12$ for a 12-bit system and $m = 1$ additional resolution bit.

The required 12-bit acquisition time in unipolar mode is approximated by [Equation 2-2](#).

$$t_{ACQ} = 9 \times (R_{MUX} + R_{MUX}) \times C_{SAMPLE}$$

For the dedicated channel (V_P/V_N), the minimum acquisition time (bipolar mode) required is given by [Equation 2-3](#).

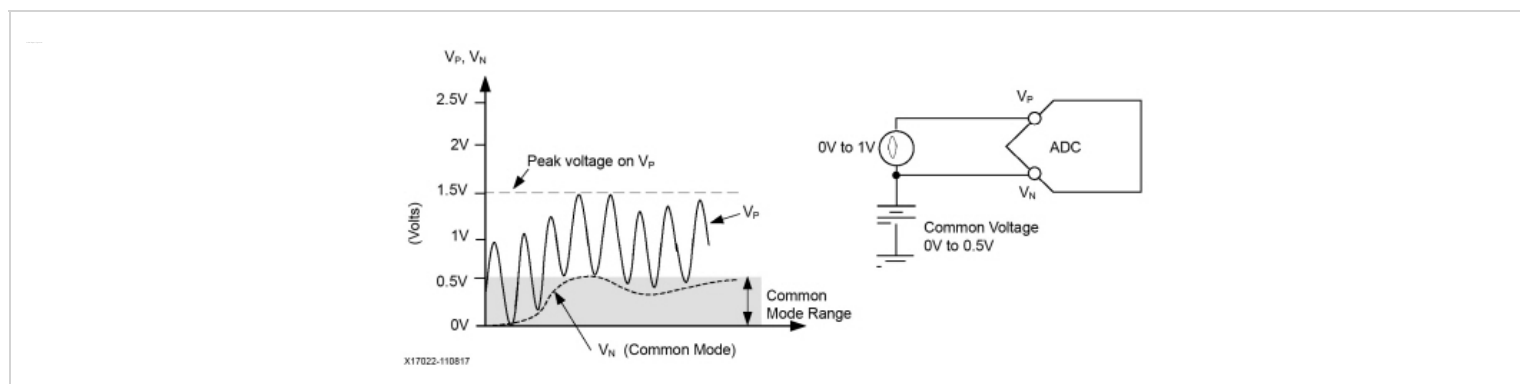
$$t_{ACQ} = 9 \times 100 \times 3 \times 10^{-12} = 3 \text{ ns}$$

The auxiliary analog channels (such as, $VAUXP[15:0]$ and $VAUXN[15:0]$) have a much larger R_{MUX} resistance that is approximately equal to $10 \text{ k}\Omega$. [Equation 2-4](#) shows the minimum acquisition time in bipolar mode.

$$t_{ACQ} = 9 \times 10 \times 10^3 \times 3 \times 10^{-12} = 300 \text{ ns}$$

Note: Any additional external resistance (for example, the anti-alias filter or resistor divider) increases the acquisition time requirement because of the increased R_{MUX} value in [Equation 2-1](#). To calculate the new acquisition time, convert any external resistance to a series equivalent resistance value and add to the R_{MUX} resistance specified in [Equation 2-3](#) and [Equation 2-4](#). For more information and design considerations for driving the ADC inputs, see XAPP795, *Driving the Xilinx Analog-to-Digital Converter* Application Note [\[Ref 7\]](#).

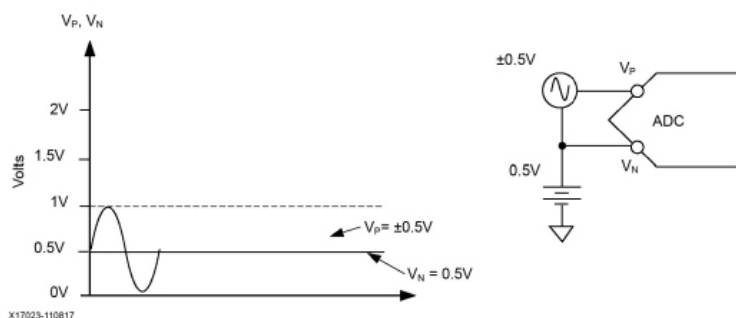
When measuring unipolar analog input signals, the ADCs must operate in a unipolar input mode. This mode is selected by writing to configuration register 0 (see [Control Registers, page 35](#)). When unipolar operation is enabled, the differential analog inputs (V_P and V_N) have an input range of 0V to 1.0V. In this mode, the voltage on V_P (measured with respect to V_N) must always be positive. [Figure 2-6](#) shows a typical application of unipolar mode. V_N is typically connected to a local ground or common mode signal. The common mode signal on V_N can vary from 0V to +0.5V (measured with respect to GNDADC). Because the differential input range is from 0V to 1.0V (V_P to V_N), the maximum signal on V_P is 1.5V. [Figure 2-6](#) shows the maximum signal levels on V_N and V_P in unipolar mode, measured with respect to analog ground (GNDADC package ball).



Unipolar Input Signal Range

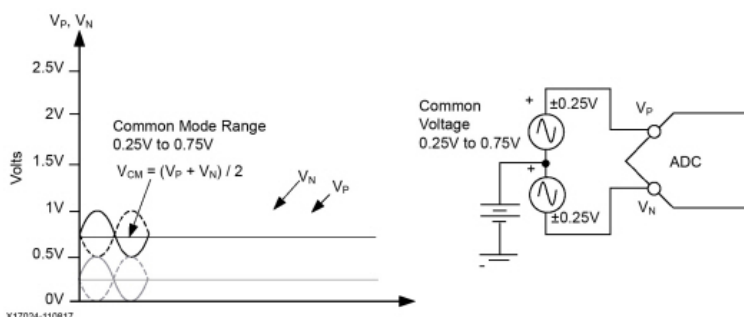
The analog inputs can accommodate analog input signals that are positive and negative with respect to a common mode or reference. To accommodate these types of signals, the analog input must be configured to bipolar mode. Bipolar mode is selected by writing to configuration register 0 (see [Control Registers](#)). All input voltages must be positive with respect to analog ground (GNDADC).

When bipolar operation is enabled, the differential analog input ($V_P - V_N$) can have a maximum input range of $\pm 0.5V$. The common mode or reference voltage should not exceed 0.5V in this case (see [Figure 2-7](#)).



Bipolar Input Signal Range

The bipolar input mode also accommodates inputs signals driven from a true differential source, for example, a balanced bridge. In this case, V_N and V_P can swing positive and negative relative to a common mode or reference voltage (see [Figure 2-8](#)). The maximum differential input ($V_P - V_N$) is $\pm 0.5V$. With maximum differential input voltages of $\pm 0.5V$ and assuming balanced inputs on V_N and V_P , the common mode voltage must lie in the range 0.25V to 0.75V.



Differential Input Signal Range

The XADC contains a temperature sensor that produces a voltage output proportional to the die temperature. [Equation 2-5](#) shows the output voltage of the temperature sensor.

$$Voltage = 10 \times \frac{kT}{q} \times \ln(10)$$

Where:

k = Boltzmann's constant = 1.38×10^{-23} J/K

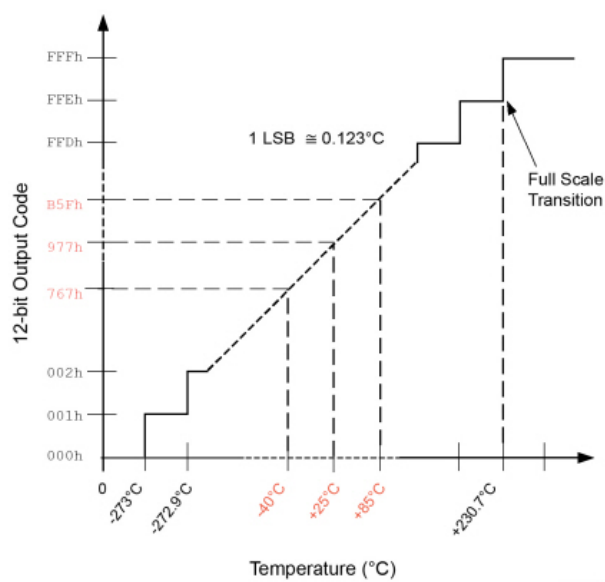
T = Temperature K (Kelvin) = $^{\circ}\text{C} + 273.15$

q = Charge on an electron = 1.6×10^{-19} C

The output voltage of this sensor is digitized by the ADC to produce a 12-bit digital output code (ADC code). [Figure 2-9](#) illustrates the digital output transfer function for this temperature sensor.

For simplification, the temperature sensor plus the ADC transfer function is rewritten as shown in [Equation 2-6](#).

$$Temperature (^{\circ}\text{C}) = \frac{ADC\ Code \times 503.975}{4096} - 273.15$$



Temperature Sensor Transfer Function

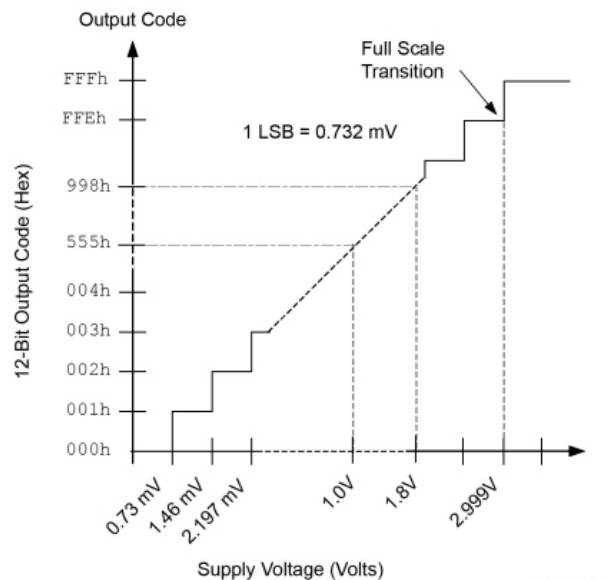
The temperature measurement result is stored in the status registers at DRP address 00h . Monitoring FPGA on-chip temperature avoids functional and irreversible failures by ensuring critical operating temperatures are not exceeded.

The XADC also includes on-chip sensors that allow a user to monitor the FPGA power-supply voltages using the ADC. The sensors sample and attenuate (by a factor of three) the power supply voltages V_{CCINT} , V_{CCAUX} , and V_{CCBRAM} on the package power supply balls. On Zynq-7000 SoC devices, the V_{CCPINT} , V_{CCPAUX} , and V_{CCO_DDR} supplies are also monitored.

Figure 2-10 shows the power-supply sensor transfer function after digitizing by the ADC. The power supply sensor can be used to measure voltages in the range 0V to $V_{CCAUX} + 5\%$ with a resolution of approximately 0.73 mV. The transfer function for the supply sensor is shown in Equation 2-7 .

$$Voltage = \frac{ADC\ Code}{4096} \times 3V$$

The power-supply measurement results for V_{CCINT} , V_{CCAUX} , and V_{CCBRAM} are stored in the status registers at DRP addresses 01h , 02h , and 06h , respectively. On Zynq-7000 SoC devices, the measurements for V_{CCPINT} , V_{CCPAUX} , and V_{CCO_DDR} are stored in status registers 0Dh , 0Eh , and 0Fh , respectively.

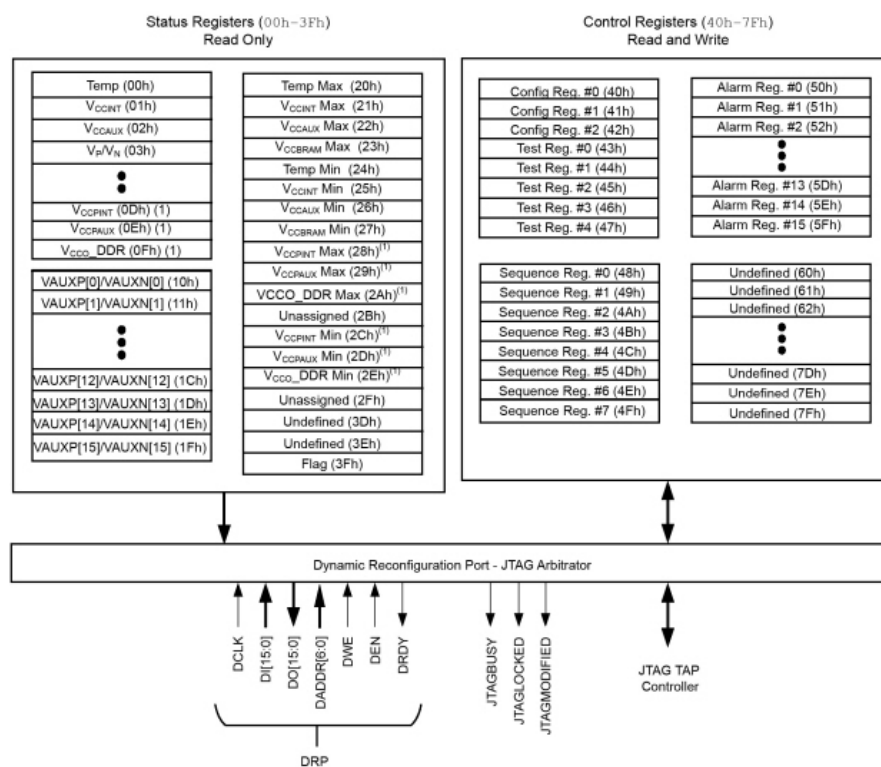


X17026-110817

Ideal Power Supply Transfer Function

XADC Register Interface

Figure 3-1 illustrates the XADC register interface. All registers in the register interface are accessible through the dynamic reconfiguration port (DRP). The DRP can be accessed through a FPGA logic port or the JTAG TAP. Access is governed by an arbitrator (see [DRP Arbitration, page 44](#)). The DRP allows you to access up to 128 16-bit registers (DADDR[6:0] = 00h to 7Fh). The first 64 access locations (DADDR[6:0] = 00h to 3Fh) are read-only and contain the ADC measurement data. These registers are known as status registers. The control registers are located at addresses 40h to 7Fh and are readable or writable through the DRP. The DRP timing is shown in [Figure 5-3](#).



X17027-110817

XADC Register Interface

Notes relevant to [Figure 3-1](#) :

1. Zynq-7000 family of SoC devices only.

For a detailed description of the DRP timing, see [Dynamic Reconfiguration Port \(DRP\) Timing, page 64](#) . For more information on the JTAG DRP interface, see [DRP JTAG Interface, page 40](#) .

Status Registers

The first 64 address locations (DADDR[6:0] = 00h to 3Fh) contain the read-only status registers. The status registers contain the results of an analog-to-digital conversion of the on-chip sensors and external analog channels. All sensors and external analog input channels have a unique channel address (see [Table 3-7, page 38](#)). The measurement result from each channel is stored in a status register with the same address on the DRP.

For example, the result from an analog-to-digital conversion on ADC multiplexer channel 0 (temperature sensor) is stored in the status register at address 00h . The result from ADC multiplexer channel 1 (V_{CCINT}) is stored at address 01h .

The status registers also store the maximum and minimum measurements recorded for the on-chip sensors from the device power-up or the last user reset of the XADC. [Table 3-1](#) defines the status registers.

Table: Status Registers (Read Only)

Name	Address	Description
Temperature	00h	The result of the on-chip temperature sensor measurement is stored in this location. The data is MSB justified in the 16-bit register. The 12 MSBs correspond to the temperature sensor transfer function shown in Figure 2-9, page 26 .
V_{CCINT}	01h	The result of the on-chip V_{CCINT} supply monitor measurement is stored at this location. The data is MSB justified in the 16-bit register. The 12 MSBs correspond to the supply sensor transfer function shown in Figure 2-10, page 27 .
V_{CCAUX}	02h	The result of the on-chip V_{CCAUX} data supply monitor measurement is stored at this location. The data is MSB justified in the 16 bit register. The 12 MSBs correspond to the supply sensor transfer function shown in Figure 2-10 .
V_P/V_N	03h	The result of a conversion on the dedicated analog input channel is stored in this register. The data is MSB justified in the 16-bit register. The 12 MSBs correspond to the transfer function shown in Figure 2-6, page 24

		or Figure 2-7, page 24 depending on analog input mode settings.
V_{REFP}	04h	The result of a conversion on the reference input V _{REFP} is stored in this register. The 12 MSBs correspond to the ADC transfer function shown in Figure 2-10 . The data is MSB justified in the 16-bit register. The supply sensor is used when measuring V _{REFP} .
V_{REFN}	05h	The result of a conversion on the reference input V _{REFN} is stored in this register. This channel is measured in bipolar mode with a two's complement output coding as shown in Figure 2-3, page 20 . By measuring in bipolar mode, small positive and negative offset around 0V (V _{REFN}) can be measured. The supply sensor is also used to measure V _{REFN} , thus 1 LSB = 3V/4096. The data is MSB justified in the 16-bit register.
V_{CCBRAM}	06h	The result of the on-chip V _{CCBRAM} supply monitor measurement is stored at this location. The data is MSB justified in the 16-bit register. The 12 MSBs correspond to the supply sensor transfer function shown in Figure 2-10 .
Undefined	07h	This location is unused and contains invalid data.
Supply A offset	08h	The calibration coefficient for the supply sensor offset using ADC A is stored at this location.
ADC A offset	09h	The calibration coefficient for the ADC A offset is stored at this location.
ADC A gain	0Ah	The calibration coefficient for the ADC A gain error is stored at this location.
Undefined	0Bh to 0Ch	These locations are unused and contain invalid data.
V_{CCPINT}⁽¹⁾	0Dh	The result of a conversion on the PS supply, V _{CCPINT} is stored in this register. The 12 MSBs correspond to the ADC transfer function shown in Figure 2-10, page 27 . The data is MSB justified in the 16-bit register. The supply sensor is used when measuring

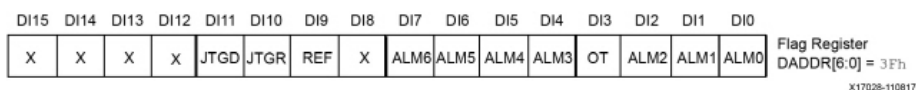
		V_{CCPINT}
$V_{CCPAUX}^{(1)}$	0Eh	The result of a conversion on the PS supply, V_{CCPAUX} is stored in this register. The 12 MSBs correspond to the ADC transfer function shown in Figure 2-10, page 27 . The data is MSB justified in the 16-bit register. The supply sensor is used when measuring V_{CCPAUX} .
$V_{CCO_DDR}^{(1)}$	0Fh	The result of a conversion on the PS supply, V_{CCO_DDR} is stored in this register. The 12 MSBs correspond to the ADC transfer function shown in Figure 2-10, page 27 . The data is MSB justified in the 16-bit register. The supply sensor is used when measuring V_{CCO_DDR} .
VAUXP[15:0]/ VAUXN[15:0]	10h to 1Fh	The results of the conversions on auxiliary analog input channels are stored in this register. The data is MSB justified in the 16-bit register. The 12 MSBs correspond to the transfer function shown in Figure 2-2, page 19 or Figure 2-3, page 20 depending on analog input mode settings.
Max temp	20h	Maximum temperature measurement recorded since power-up or the last XADC reset.
Max V_{CCINT}	21h	Maximum V_{CCINT} measurement recorded since power-up or the last XADC reset.
Max V_{CCAUX}	22h	Maximum V_{CCAUX} measurement recorded since power-up or the last XADC reset.
Max V_{CCBRAM}	23h	Maximum V_{CCBRAM} measurement recorded since power-up or the last XADC reset.
Min temp	24h	Minimum temperature measurement recorded since power-up or the last XADC reset.
Min V_{CCINT}	25h	Minimum V_{CCINT} measurement recorded since power-up or the last XADC reset.
Min V_{CCAUX}	26h	Minimum V_{CCAUX} measurement

		recorded since power-up or the last XADC reset.
Min V_{CCBRAM}	27h	Minimum V _{CCBRAM} measurement recorded since power-up or the last XADC reset.
V_{CCPINT}⁽¹⁾ max	28h	Maximum V _{CCPINT} measurement recorded since power-up or the last XADC reset.
V_{CCPAUX}⁽¹⁾ max	29h	Maximum V _{CCPAUX} measurement recorded since power-up or the last XADC reset.
V_{CCO_DDR}⁽¹⁾ max	2Ah	Maximum V _{CCO_DDR} measurement recorded since power-up or the last XADC reset.
Unassigned	2Bh	
V_{CCPINT}⁽¹⁾ min	2Ch	Minimum V _{CCPINT} measurement recorded since power-up or the last XADC reset.
V_{CCPAUX}⁽¹⁾ min	2Dh	Minimum V _{CCAUX} measurement recorded since power-up or the last XADC reset.
V_{CCO_DDR}⁽¹⁾ min	2Eh	Minimum V _{CCO_DDR} measurement recorded since power-up or the last XADC reset.
Unassigned	2Fh	
Supply B offset	30h	The calibration coefficient for the supply sensor offset using ADC B is stored at this location.
ADC B offset	31h	The calibration coefficient for the ADC B offset is stored at this location.
ADC B gain	32h	The calibration coefficient for the ADC B gain error is stored at this location.
Undefined	33h to 3Eh	These locations are unused and contain invalid data.
Flag	3Fh	This register contains general status information (see Flag Register).

Notes:

1. These channels are only available in Zynq-7000 SoC devices.

The Flag Register is shown in [Figure 3-2](#). The bit definitions are described in [Table 3-2](#).



Flag Registers

Table: Flag Register Bit Definitions

Bit	Name	Description
D17 to D10	ALM6 to ALM0	These bits reflect the status of the alarm output ALM[6:0].
D13	OT	This bit reflects the status of the over temperature logic output.
D19	REF	When this bit is a logic 1, the ADC is using the internal voltage reference. When this bit is a logic 0, the external reference is being used.
D110	JTGR	A logic 1 indicates that the JTAG_XADC bitstream option has been used to restrict JTAG access to read only. See DRP JTAG Interface for more information.
D111	JTGD	A logic 1 indicates that the JTAG_XADC bitstream option has been used to disable all JTAG access. See DRP JTAG Interface for more information.

The XADC can digitally calibrate out any offset and gain errors in the ADCs and power supply sensor. By connecting known voltages (for example, V_{REFP} and V_{REFN} as opposed to the internal reference) to the ADCs and the supply sensor, the offset and gain errors can be calculated and correction coefficients generated. These calibration coefficients are stored in status registers 08h to 0Ah for ADC A and 30h to 32h for ADC B (see [Table 3-1, page 30](#)).

The XADC has a built-in calibration function that automatically calculates these coefficients. By initiating a conversion on channel 8 (08h), all calibration coefficients are calculated. The XADC *default* operating mode automatically uses calibration. When not operating in the default mode, these calibration coefficients are applied to all ADC measurements by enabling the calibration bits (CAL0–3) in configuration register 1 (41h) (see [Table 3-5, page 37](#)).

BUSY transitions High for the duration of the entire calibration sequence (conversion on channel 8). This calibration sequence is four times longer than a regular conversion on a sensor channel as offset and gain are measured for both ADCs and the power supply sensor.

As mentioned previously, the offset and gain calibration coefficients are stored in the status registers. This section explains how to interpret the values in these registers. These are read-only registers, and it is not possible to modify the contents through the DRP. See [Figure 3-3](#).

DI15	DI14	DI13	DI12	DI11	DI10	DI9	DI8	DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0
DATA[11:0]												Note		ADC A Supply Offset (08h)	
DATA[11:0]												Note			
N/A								Sign	MAG[5:0]					ADC A Gain (0Ah)	
DATA[11:0]												Note			
DATA[11:0]												Note		ADC B Supply Offset (30h)	
N/A								Sign	MAG[5:0]						
DATA[11:0]												Note		ADC B Bipolar Offset (31h)	
N/A								Sign	MAG[5:0]						
DATA[11:0]												Note		ADC B Gain (32h)	
N/A								Sign	MAG[5:0]						

X17029-110817

Calibration Coefficients in Status Register

Note: The ADCs always produce a 16-bit conversion result. The 12-bit data correspond to the 12 MSBs (most significant) in the 16-bit status registers. The unreferenced LSBs can be used to minimize quantization effects or improve resolution through averaging or filtering.

The offset calibration registers store the offset correction factor for the supply sensor and ADC. The offset correction factor is a 12-bit, two's complement number and is expressed in LSBs. Similar to other status registers, the 12-bit values are MSB justified in the registers. For example, if the ADC has an offset of +10 LSBs (approximately $10 \times 250 \mu\text{V} = 2.5 \text{ mV}$), the offset coefficient records -10 LSBs or FF6h, and status register 08h records 1111, 1111, 1100 XXXXb. For the supply sensor, the LSB size is approximately $750 \mu\text{V}$, thus a +10 LSB offset is equivalent to 7.5 mV of offset in the supply measurement.

The ADC gain calibration coefficient stores the correction factor for any gain error in the ADCs. The correction factor is stored in the seven LSBs of registers 0Ah and 32h. These seven bits store both sign and magnitude information for the gain correction factor. If the seventh bit is a logic 1, the correction factor is positive. If it is 0, the correction factor is negative. The next six bits store the magnitude of the gain correction factor. Each bit is equivalent to 0.1%.

For example, if the ADC A has a positive gain error of +1%, then the gain calibration coefficient records -1% (the -1% correction applied to cancel the +1% error). Because the correction factor is negative, the seventh bit is set to zero. The remaining magnitude bits record 1%, where $1\% = 10 \times 0.1\%$ and $10 = 001010$ binary. The status register 0Ah records 0000 0000 0000 1010. With six bits assigned to the magnitude and a maximum value of 3Fh, the calibration can correct errors in the range of $\pm 0.1\% \times 63$, or 6.3%.

Control Registers

The XADC has 32 control registers that are located at DRP addresses 40h to 5Fh (see [Table 3-3](#)). These registers are used to configure the XADC operation. All XADC functionality is controlled through these registers.

These control registers are initialized using the XADC attributes when the XADC is instantiated in a design. This means that the XADC can be configured to start in a predefined mode after FPGA configuration.

Table: XADC Control Registers

Name	Address	Software Attribute	Description
Configuration register 0	40h	INIT_40	These are XADC configuration registers (see Configuration Registers (40h to 42h)).
Configuration register 1	41h	INIT_41	
Configuration register 2	42h	INIT_42	
Test registers 0 to 4	43h to 47h	INIT_43 to INIT_47	These are test registers. The default initialization is 0000h. These registers are used for factory test and should be left at the default initialization.

Sequence registers	48h to 4Fh	INIT_48 to INIT_4F	These registers are used to program the channel sequencer function (see Chapter 4, XADC Operating Modes).
Alarm registers	50h to 5Fh	INIT_50 to INIT_5F	These are the alarm threshold registers for the XADC alarm function (see Automatic Alarms, page 56).

The first three registers in the control register block configure the XADC operating modes. These registers are known as XADC configuration registers. Their bit definitions are illustrated in [Figure 3-4](#) .

Note: Bits shown as 0 should always be left set to 0 .

The configuration registers can be modified through the DRP after the FPGA has been configured. For example, a soft microprocessor or state machine can be used to alter the contents of the XADC control registers at any time during normal operation. [Table 3-4](#) through [Table 3-6](#) define the bits for the three configuration registers.

DI15	DI14	DI13	DI12	DI11	DI10	DI9	DI8	DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0	Config Reg #0 DADDR[6:0] = 40h
CAVG	0	AVG1	AVG0	MUX	B $\bar{U}$	E $\bar{C}$	ACQ	0	0	0	CH4	CH3	CH2	CH1	CH0	
DI15	DI14	DI13	DI12	DI11	DI10	DI9	DI8	DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0	Config Reg #1 DADDR[6:0] = 41h
SEQ3	SEQ2	SEQ1	SEQ0	ALM6 (Note1)	ALM5 (Note1)	ALM4 (Note1)	ALM3	CAL3	CAL2	CAL1	CAL0	ALM2	ALM1	ALM0	OT	
DI15	DI14	DI13	DI12	DI11	DI10	DI9	DI8	DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0	Config Reg #2 DADDR[6:0] = 42h
CD7	CD6	CD5	CD4	CD3	CD2	CD1	CD0	0	0	PD1	PD0	0	0	0	0	X17030-110817

Configuration Registers Bit Definitions

Notes relevant to [Figure 3-4](#) :

1. Zynq-7000 SoC devices only.

Table: Configuration Register 0 Bit Definitions

Bit	Name	Description
DI0 to DI4	CH0 to CH4	When operating in single channel mode or external multiplexer mode, these bits are used to select the ADC input channel. See Table 3-7 for the channel assignments.
DI8	ACQ	When using single channel mode, this bit is used to increase the settling time available on external analog inputs in continuous sampling mode by six ADCCLK cycles (see Chapter 2, Analog-to-Digital Converter and Chapter 5, XADC Timing). The acquisition time is increased by setting this bit to logic 1. (1)
DI9	E C	This bit is used to select either

		continuous or event-driven sampling mode for the ADC (see Chapter 5, XADC Timing). A logic 1 places the ADC in event-driven sampling mode and a logic 0 places the ADC in continuous sampling mode.
DI10	B U	This bit is used in single channel mode to select either unipolar or bipolar operating mode for the ADC analog inputs (see Analog Inputs, page 21). A logic 1 places the ADC in bipolar mode and a logic 0 places the ADC in unipolar mode.
DI11	MUX	This bit should be set to a logic 1 to enable external multiplexer mode. See External Multiplexer Mode for more information.
DI12, DI13	AVG0, AVG1	These bits are used to set the amount of sample averaging on selected channels in both single channel and sequence modes (see Table 3-8 and Chapter 4, XADC Operating Modes).
DI15	CAVG	This bit is used to disable averaging for the calculation of the calibration coefficients. Averaging is enabled by default (logic 0). To disable averaging, set this bit to logic 1. Averaging is fixed at 16 samples.

Notes:

1. Acquisition times for sequencer modes use channel sequencer registers; see [ADC Channel Settling Time \(4Eh and 4Fh\)](#) in [Chapter 4](#).

Table: Configuration Register 1 Bit Definitions

Bit	Name	Description
DI0	OT	This bit is used to disable the over-temperature signal. The alarm is disabled by setting this bit to logic 1.
DI1 to DI3, DI8	ALM0 to ALM3	These bits are used to disable individual alarm outputs for temperature, V_{CCINT} , $V_{CCAU X}$, and V_{CCBRAM} , respectively. A logic 1 disables an alarm output.
DI9 to DI11	ALM4 to ALM6	These bits are used to disable individual alarm outputs for V_{CCPINT} , V_{CCPAUX} , and V_{CCO_DDR} , respectively. A logic 1 disables an alarm

		output.
DI4 to DI7	CAL0 to CAL3	These bits enable the application of the calibration coefficients to the ADC and on-chip supply sensor measurements. A logic 1 enables calibration and a logic 0 disables calibration. For bit assignments, see Table 3-10 .
DI12 to DI15	SEQ0 to SEQ3	These bits enable the channel-sequencer function. For the bit assignments, see Table 3-9 . See Chapter 4, XADC Operating Modes , for more information.

Table: Configuration Register 2 Bit Definitions

Bit	Name	Description
DI4, DI5	PD0, PD1	Power-down bits for the XADC. The entire XADC block can be powered down permanently by setting PD1 = PD0 = 1. ADC B can also be powered down permanently by setting PD1 = 1 and PD0 = 0. See Table 3-11 .
DI8 to DI15	CD0 to CD7	These bits select the division ratio between the DRP clock (DCLK) and the lower frequency ADC clock (ADCCLK) used for the ADC (see Chapter 5, XADC Timing). For bit assignments, see Table 3-12 .

Table: ADC Channel Select

ADC Channel	CH4	CH3	CH2	CH1	CH0	Description
0	0	0	0	0	0	On-chip temperature
1	0	0	0	0	1	V _{CCINT}
2	0	0	0	1	0	V _{CCAUX}
3	0	0	0	1	1	V _P , V _N —Dedicated analog inputs
4	0	0	1	0	0	V _{REFP} (1.25V) ⁽¹⁾
5	0	0	1	0	1	V _{REFN} (0V) ⁽¹⁾
6	0	0	1	1	0	V _{CCBRAM}
7	0	0	1	1	1	Invalid channel

						selection
8	0	1	0	0	0	Carry out an XADC calibration
9–12	...	...	...	...	...	Invalid channel selection
13	0	1	1	0	1	V _{CCPINT} ⁽³⁾
14	0	1	1	1	0	V _{CCPAUX} ⁽³⁾
15	0	1	1	1	1	V _{CCO_DDR} ⁽³⁾
16	1	0	0	0	0	VAUXP[0], VAUXN[0] – Auxiliary channel 0
17	1	0	0	0	1	VAUXP[1], VAUXN[1] – Auxiliary channel 1
18–31	...	...	...	...	...	VAUXP[2:15], VAUXN[2:15] – Auxiliary channels 2 to 15 ⁽²⁾

Notes:

1. These channel selection options are used for XADC self-check and calibration operations. When these channels are selected, the supply sensor is connected to V_{REFP} or V_{REFN}.
2. Auxiliary channels 6, 7, 13, 14, and 15 are not supported on Kintex® -7 devices. Some auxiliary analog channels might also not be supported in certain Virtex-7, Artix-7, Spartan-7, and Zynq-7000 SoC package options. Users should consult the package file for the device.
3. These channels are only supported in Zynq-7000 SoC devices.

Table: Averaging Filter Settings

AVG1	AVG0	Function
0	0	No averaging
0	1	Average 16 samples
1	0	Average 64 samples
1	1	Average 256 samples

Table: Sequencer Operation Settings

SEQ3	SEQ2	SEQ1	SEQ0	Function
0	0	0	0	Default mode

0	0	0	1	Single pass sequence
0	0	1	0	Continuous sequence mode
0	0	1	1	Single channel mode (sequencer off)
0	1	X	X	Simultaneous sampling mode
1	0	X	X	Independent ADC mode
1	1	X	X	Default mode

Table: Calibration Enables

Name	Description
CAL0	ADCs offset correction enable
CAL1	ADCs offset and gain correction enable
CAL2	Supply sensor offset correction enable
CAL3	Supply sensor offset and gain correction enable

Table: Power Down Selection

PD1	PD0	Description
0	0	Default. All XADC blocks powered up
0	1	Not valid – do not select
1	0	ADC B powered down
1	1	XADC powered down

Table: DCLK Division Selection ⁽¹⁾ ⁽²⁾

CD7	CD6	CD5	CD4	CD3	CD2	CD1	CD0	Division
0	0	0	0	0	0	0	0	2
0	0	0	0	0	0	0	1	2
0	0	0	0	0	0	1	0	2
0	0	0	0	0	0	1	1	3
0	0	0	0	0	1	0	0	4
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
1	1	1	1	1	1	1	0	254

1	1	1	1	1	1	1	1	255
---	---	---	---	---	---	---	---	-----

Notes:

1. Minimum division ratio is 2, for example, $ADCCLK = DCLK/2$.
2. DCLK division must be selected to keep the ADC clock in its supported frequency range as specified in the applicable data sheet. To find the data sheet, see the [Xilinx website](#).

These registers, intended for factory test purposes only, have a default status of zero. You must not write to these registers.

These registers are used to program the channel sequencer functionality. For more information see [Automatic Channel Sequencer, page 46](#).

These registers are used to program the alarm thresholds for the automatic alarms on the internally monitored channels, temperature, V_{CCINT} , V_{CCAUX} , and V_{CCBRAM} . For Zynq-7000 SoC devices, the alarm thresholds for V_{CCPINT} , V_{CCPAUX} , and V_{CCO_DDR} are also set using these registers. For more information, see [Automatic Alarms, page 56](#).

DRP JTAG Interface

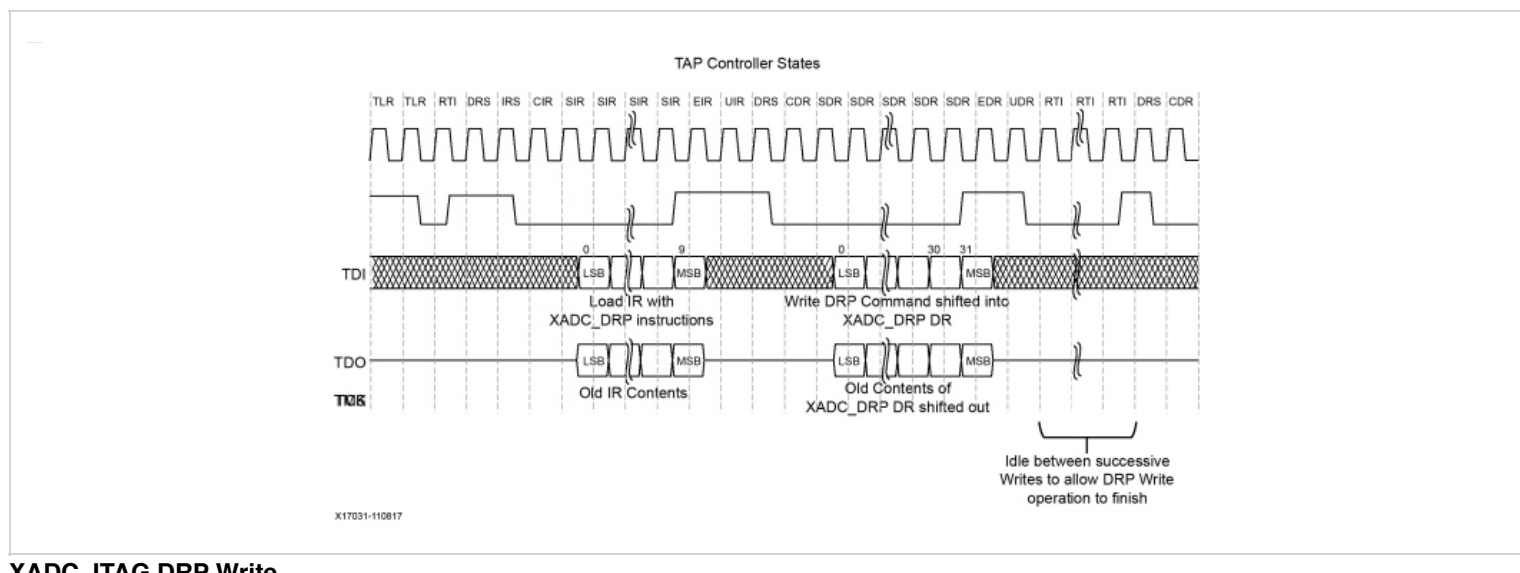
The XADC uses a full JTAG interface extension to the DRP interface. This allows read/write access to the XADC DRP through the existing on-chip JTAG infrastructure. No instantiation is required to access the DRP interface over JTAG. A boundary-scan instruction (6-bit instruction = 110111) called XADC_DRP, added to 7 series FPGAs, allows access to the DRP through the JTAG TAP. All XADC JTAG instructions are 32 bits wide. For more information on the 7 series FPGA boundary-scan instructions and usage, see UG470, *7 Series FPGAs Configuration User Guide* [Ref 3]. Read and Write operations using the XADC JTAG DRP interface are described in the next sections. Users unfamiliar with basic JTAG functionality should become familiar with the JTAG standard (IEEE 1149.1) before proceeding.

Important : In ISE tools, JTAG access can be limited to read only or completely disabled by using the **JTAG_XADC: [Enable | Disable | Status_only]** BitGen option. If the dedicated PS to XADC interface is used in the Zynq-7000 SoC devices, this external JTAG access is also disabled (see UG585, *Zynq-7000 All Programmable SoC Technical Reference Manual* [Ref 4] for more details). When using Vivado tools, JTAG access can be adjusted by adding the following to an XDC file:

```
set_property BITSTREAM.GENERAL.JTAG_XADC <Enable|Disable|StatusOnly> current_design_name
```

See UG908, *Vivado Design Suite User Guide: Programming and Debugging* [Ref 8] for more information on device configuration bitstream settings.

Figure 3-5 shows a timing diagram for a write operation to the XADC DRP through the JTAG TAP. The DRP is accessed through the XADC data register (XADC DR). Before the XADC DR is accessed, the instruction register (IR) must first be loaded with the XADC instruction. The controller is placed in the IR-scan mode, and the XADC instruction is shifted to the IR.



XADC JTAG DRP Write

Page 33 of 69

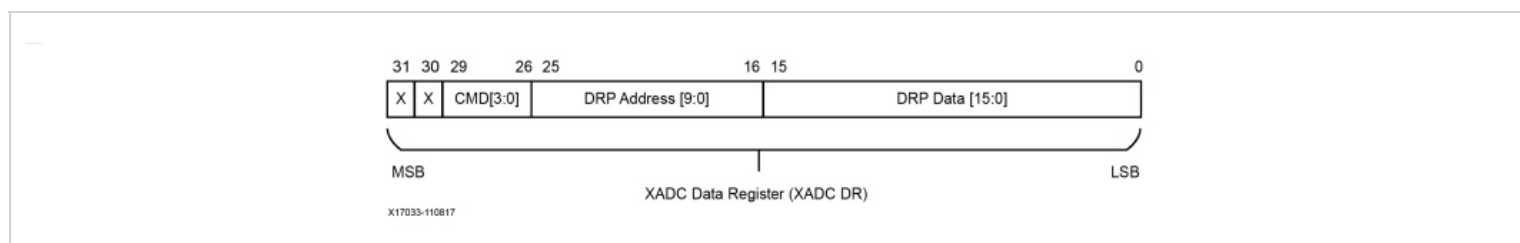
During the DR-Capture phase of the next DR-scan operation, newly read data is transferred from the arbitrator to the XADC DR. This 16-bit data (stored in the 16 LSBs of the 32-bit word) is then shifted out on TDO during the subsequent shift operation (see [Figure 3-6](#)). The timing diagram shows several idle states at the end of the first DR-scan operation, allowing the arbitrator enough time to fetch the XADC DRP data.

As mentioned previously, if the DCLK frequency is significantly faster than the TCK, these idle states might not be required.

Implementing a DR-scan operation before the arbitrator has completed the DRP-read operation results in old DRP data being transferred to the XADC DR during the DR-capture phase.

To ensure reliable operation over all operating clock frequencies, a minimum of 10 Run-Test-Idle (RTI) states should be inserted. Multiple read operations can be pipelined, as shown in [Figure 3-6](#). Thus, as the result of a read operation is being shifted out of the XADC DR, an instruction for the next read can be shifted in.

The data shifted into the 32-bit XADC DR during a DR-scan operation instructs the arbitrator to carry out a write, read, or no operation on the XADC DRP. [Figure 3-7](#) shows the data format of the JTAG DRP command loaded into the XADC DR. The first 16 LSBs of XADC DR [15:0] contain the DRP register data. For both read and write operations, the address bits XADC DR [25:16] hold the DRP target register address. The command bits XADC DR [29:26] specify a read, write, or no operation (see [Table 3-13](#)).



XADC JTAG DRP Command

Table: JTAG DRP Commands

CMD[3:0]				Operation
0	0	0	0	No operation
0	0	0	1	DRP Read
0	0	1	0	DRP Write
-	-	-	-	Not defined

It is also possible to enable the auxiliary analog input channel preconfiguration of the FPGA, allowing external analog voltages (on the PCB) to be monitored using the JTAG TAP before configuration. The auxiliary channels are enabled by writing 0001h to DRP address 02h. This address lies within the read-only status register address space and normally holds the result of a V_{CCAUX} measurement. However, a write to this address enables the auxiliary inputs. This function only works prior to configuration. After configuration, these inputs must be explicitly instantiated in the design.

Because the DRP registers are accessed from two different ports (interconnect and JTAG TAP), access must be carefully managed. An arbitrator is implemented to manage potential conflicts between the FPGA logic (fabric) and JTAG port. Arbitration is managed on a per transaction basis (a transaction is a single read/write operation to the DRP). The arbitration rules are as follows:

1. A JTAG transaction cannot be interrupted by the FPGA logic. The FPGA logic transaction is queued by the arbitrator until the JTAG transaction has finished, and then the FPGA logic transaction is completed.
2. A JTAG transaction cannot interrupt an FPGA logic transaction already in progress. As soon as the FPGA logic transaction is finished, then the JTAG transaction is completed.

Three status signals help manage access through the interconnect when the JTAG port is also being used: JTAGBUSY,

JTAGMODIFIED, and JTAGLOCKED.

JTAGBUSY becomes active during the update phase of a DRP transaction through the JTAG TAP. This signal resets when the JTAG XADC DR transaction is completed. Each read/write to the XADC DR is treated as an individual transaction. If DRP access initiates through the interconnect port when JTAGBUSY is High, then the arbitrator queues this request for a read/write through the FPGA logic. DRDY does not go active until JTAGBUSY transitions Low and the interconnect transaction is completed. A second DRP access through the FPGA logic must not be initiated until the DRDY for the initial access becomes active and indicates the read/write was successful. If an interconnect access is in progress when a JTAG DRP transaction initiates, the interconnect access is completed before the JTAG transaction.

Whenever there is a JTAG write (JTAG reads typically occur more often) to any register in the DRP, the application (FPGA) must be notified about the potential change of configuration. Thus, the JTAGMODIFIED signal transitions High after a JTAG write. A subsequent DRP read/write resets the signal.

In some cases, it is simpler for the JTAG user to take DRP ownership for a period by locking out access through the interconnect. This is useful in a diagnostic situation where a large number of DRP registers are modified through the JTAG TAP. When a JTAGLOCKED request is made, the JTAGLOCKED signal transitions to the active-High state. The signal remains High until the port is unlocked again. No read or write access is possible through the DRP logic port when the JTAGLOCKED signal is High. The JTAGLOCKED signal is activated by writing 0001h to DRP address 00h . The JTAGLOCKED signal is reset by writing 0000h to DRP address 00h .

JTAGLOCKED is also used to indicate when the DRP is ready for a read or write when the DCLK is first connected or when DCLK becomes active again after a period of inactivity. It can take up to 10 DCLK cycles for JTAGLOCKED to go Low after DCLK becomes active.

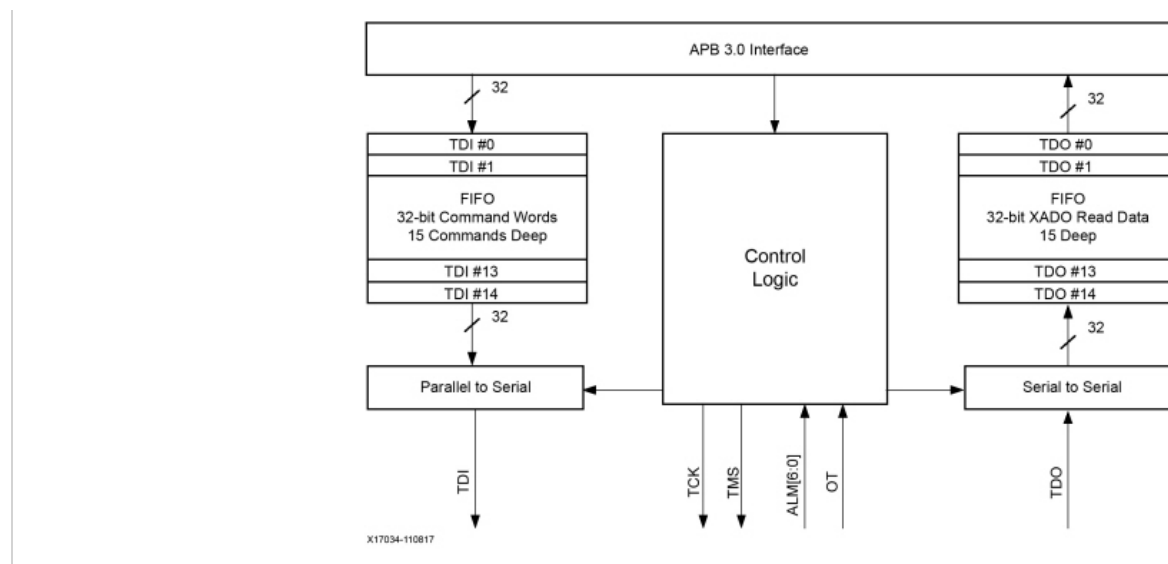
Note: The XADC automatically switches over to an on-chip clock oscillator if a missing DCLK is detected.

A user reset of the XADC can also be initiated using the JTAG interface. The XADC is reset by writing xxxh (any 16-bit value) to DRP address 03h . The JTAG reset has the same effect as pulsing the RESET pin.

Zynq-7000 SoC Processing System (PS) to XADC Dedicated Interface

The DRP JTAG interface described in [DRP JTAG Interface, page 40](#) is also used to provide a dedicated interface between the processor subsystem and the XADC block located in the programmable logic region of the Zynq-7000 SoC. This dedicated interface does not require the programmable logic to be configured. All the XADC functionality can be accessed by writing to and reading from the XADC status and configuration registers. For more details on how to enable this interface and programming models, see UG585, *Zynq-7000 All Programmable SoC Technical Reference Manual* [\[Ref 4\]](#) .

Note: When this interface is enabled, external JTAG access to the XADC as described in [DRP JTAG Interface, page 40](#) is no longer supported.



Processor to XADC Dedicated Interface

Figure 3-8 illustrates a simplified block diagram of the interface between the processor subsystem and the XADC block. To simplify the interface and also reduce the overhead on the processor, two 32-bit FIFOs have been implemented. Up to 15 32-bit DRP JTAG command words (see Table 3-13) can be loaded into the command FIFO. The control logic in the interface manages the parallel to serial conversion and writing the commands to the DRP JTAG interface. Serial data shifted out on TDO on the DRP JTAG interface is converted to a 32-bit parallel word and written to the DRP JTAG Read Data FIFO which is also 15 words deep. XADC read data is accessed by the processor by reading from this FIFO. The same timing and protocol as detailed in DRP JTAG Interface, page 40, applies to this interface.

The XADC alarm signals OT and ALM[6:0] are also connected to this dedicated interface. These signals generate interrupts when the XADC automatic alarms go active. For more information, see Automatic Alarms, page 56.

XADC Operating Modes

The XADC includes several operating modes that cover some of the most common use cases for this kind of functionality. The most basic mode of operation is called *default mode*, where the XADC monitors all on-chip sensors and requires no configuration of the XADC. In the *simultaneous sampling mode*, the sequencer is used to operate both ADCs in lock step to sample two external analog inputs and store results in the status registers.

The XADC operating modes can be used with either continuous or event-driven sampling modes unless otherwise noted.

Single Channel Mode

This mode is enabled when bits SEQ3 to SEQ0 in control register 41h are set to 0011 (see Table 3-9, page 39). In this mode, you must select the channel for analog-to-digital conversion by writing to bit locations CH4 to CH0 in control register 40h. Various configurations for single channel mode, such as analog input mode (B U) and settling time (ACQ), must also be set by writing to control register 40h. In applications where many channels need to be monitored, there can be a significant overhead for the microprocessor or other controller. To automate this task, a function called the automatic channel sequencer is provided.

Automatic Channel Sequencer

The automatic channel sequencer sets up a range of predefined operating modes, where several channels (on-chip sensors and external inputs) are used. The sequencer automatically selects the next channel for conversion, sets the averaging, configures the analog input channels, sets the required settling time for acquisition, and stores the results in the status registers based on a once off setting. The sequencer modes are set by writing to the SEQ3, SEQ2, SEQ1, and SEQ0 bits in configuration register 1.

(see [Table 3-9](#)).

The channel sequencer functionality is implemented using eight control registers from addresses 48h to 4Fh on the DRP (see [Control Registers, page 35](#)). These eight registers can be viewed as four pairs of 16-bit registers. Each pair of registers controls one aspect of the sequencer functionality. Individual bits in each pair of registers (32 bits) enable specific functionality for a particular ADC channel. The four pairs of registers are:

- [ADC Channel Selection Registers \(48h and 49h \)](#)
- [ADC Channel Averaging \(4Ah and 4Bh \)](#)
- [ADC Channel Analog-Input Mode \(4Ch and 4Dh \)](#)
- [ADC Channel Settling Time \(4Eh and 4Fh \)](#)

The ADC channel selection registers enable and disable a channel in the automatic channel sequencer. The bits for these registers are defined in [Table 4-1](#) and [Table 4-2](#) . The two 16-bit registers are used to enable or disable the associated channels. A logic 1 enables a particular channel in the sequence. The sequence order is fixed starting from the LSB (bit 0) of register 48h and ending with the MSB (bit 15) of register 49h . The behavior of the channel sequencer does change for simultaneous sampling mode and independent ADC mode (see [Simultaneous Sampling Mode](#) and [Independent ADC Mode](#) , respectively).

Table: Sequencer On-Chip Channel Selection (48h)

Sequence Number 7 Series/Zynq-7000	Bit	ADC Channel	Description
1/1	0	8	XADC calibration
-	1	9	Invalid channel selection
	2	10	
	3	11	
	4	12	
-/2	5	13	V _{CCPINT}
-/3	6	14	V _{CCPAUX}
-/4	7	15	V _{CCO_DDR}
2/5	8	0	On-chip temperature
3/6	9	1	V _{CCINT}
4/7	10	2	V _{CCAUX}
5/8	11	3	V _P , V _N – Dedicated analog inputs
6/9	12	4	V _{REFP}
7/10	13	5	V _{REFN}
8/11	14	6	V _{CCBRAM}
-	15	7	Invalid channel selection

Table: Sequencer Auxiliary Channel Selection (49h)

Sequence Number 7 Series/Zynq-7000	Bit	ADC Channel	Description
9/12	0	16	VAUXP[0], VAUXN[0] – Auxiliary channel
10/13	1	17	VAUXP[1], VAUXN[1] – Auxiliary channel
11/14	2	18	VAUXP[2], VAUXN[2] – Auxiliary channel
12/15	3	19	VAUXP[3], VAUXN[3] – Auxiliary channel
13/16	4	20	VAUXP[4], VAUXN[4] – Auxiliary channel
14/17	5	21	VAUXP[5], VAUXN[5] – Auxiliary channel
15/18	6	22	VAUXP[6], VAUXN[6] – Auxiliary channel
16/19	7	23	VAUXP[7], VAUXN[7] – Auxiliary channel
17/20	8	24	VAUXP[8], VAUXN[8] – Auxiliary channel
18/21	9	25	VAUXP[9], VAUXN[9] – Auxiliary channel
19/22	10	26	VAUXP[10], VAUXN[10] – Auxiliary channel
20/23	11	27	VAUXP[11], VAUXN[11] – Auxiliary channel
21/24	12	28	VAUXP[12], VAUXN[12] – Auxiliary channel
22/25	13	29	VAUXP[13], VAUXN[13] – Auxiliary channel
23/26	14	30	VAUXP[14], VAUXN[14] – Auxiliary channel
24/27	15	31	VAUXP[15], VAUXN[15] – Auxiliary channel

Notes:

1. Auxiliary channels 6, 7, 13, 14, and 15 are not supported on Kintex-7 devices. Some auxiliary analog channels might also not be supported in certain Virtex-7, Artix-7, Spartan-7, and Zynq-7000 SoC device package options. Users should consult the package file for the device.

The ADC channel averaging registers enable and disable the averaging of the channel data in a sequence. The result of a measurement on an averaged channel is generated by using 16, 64, or 256 samples. The amount of averaging is selected by using the AVG1 and AVG0 bits in configuration register 0 (see [Control Registers, page 35](#)). These registers also have the same bit assignments as the channel sequence registers listed in [Table 4-1](#) and [Table 4-2](#).

Averaging can be selected independently for each channel in the sequence. When averaging is enabled for some of the channels of the sequence, the EOS is only pulsed after the sequence has completed the amount of averaging selected by using AVG1 and AVG0 bits (see [Table 3-8, page 38](#)). If a channel in the sequence does not have averaging enabled, its status register is updated for every pass through the sequencer. When a channel has averaging enabled, its status register is only updated after the averaging is complete. An example sequence is Temperature and $V_{AUX}[1]$, where an averaging of 16 is enabled on $V_{AUX}[1]$. The sequence is Temperature, $V_{AUX}[1]$, Temperature, $V_{AUX}[1]$, ..., Temperature, $V_{AUX}[1]$ for each of the conversions where the temperature status register is updated. The $V_{AUX}[1]$ status register is updated after the averaging of the 16 conversions.

If averaging is enabled for the calibration channel by setting CAVG to a logic 0 (see [Control Registers, page 35](#)), the coefficients are updated after the first pass through the sequence. Subsequent updates to coefficient registers require 16 conversions before the coefficients are updated. Averaging is fixed at 16 samples for calibration.

These registers are used to configure an ADC channel as either unipolar or bipolar in the automatic sequence (see [Analog Inputs, page 21](#)). These registers also have the same bit assignments as the channel sequence registers listed in [Table 4-1](#) and [Table 4-2](#). However, only external analog input channels, such as the dedicated input channels (V_P and V_N) and the auxiliary analog inputs (VAUXP[15:0] and VAUXN[15:0]) can be configured in this way. Setting a bit to logic 1 enables a bipolar input mode for the associated channel. Setting a bit to logic 0 (default) enables a unipolar input mode. All internal sensors use a unipolar transfer function.

The default settling time for an external channel in continuous sampling mode is four ADCCLK cycles. The settling time is additional acquisition time after the end of a conversion (see [Chapter 5, XADC Timing](#)). However, by setting the corresponding bits (for external channels) to logic 1 in registers 4Eh and 4Fh, the associated channel can have its settling time extended to 10 ADCCLK cycles. The bit definitions (which bits correspond to which external channels) for these registers are the same as the sequencer channel selection shown in [Table 4-1](#) and [Table 4-2](#).

Sequencer Modes

There are several sequencer modes as defined by [Table 3-9, page 39](#). These modes are described in this section.

The default mode is enabled by setting the sequence bits to 0. In this mode of operation, the XADC automatically monitors the on-chip sensors and stores the results in the status registers. Both ADCs are calibrated in this mode and an averaging of 16 samples is applied to all sensors. The XADC operates independently of any other control register settings in this mode. The XADC also operates in default mode after initial power-up and during FPGA configuration. [Table 4-3](#) shows the default sequence for the XADC.

 **Note:** All alarm outputs (ALM[7:0]) except OT are disabled in default mode. ADC calibration is automatically enabled in default mode.

Table: Default Mode Sequence

Order	Channel	Address	Description
1	Calibration	08h	Calibration of ADC A and ADC B
2 (1)	V_{CCPINT}	0Dh	V_{CCPINT} supply sensor
3 (1)	V_{CCPAUX}	0Eh	V_{CCPAUX} supply sensor
4 (1)	V_{CCO_DDR}	0Fh	V_{CCO_DDR} supply sensor
2/5 (1)	Temp	00h	Temperature sensor
3/6 (1)	V_{CCINT}	01h	V_{CCINT} supply sensor
4/7 (1)	V_{CCAUX}	02h	V_{CCAUX} supply sensor
5/8 (1)	V_{CCBRAM}	06h	V_{CCBRAM} supply sensor

Notes:

1. Only available on the Zynq-7000 SoC devices.

In single pass mode, the sequencer operates for one pass through the sequencer channel select registers (48h and 49h) and then halts. A sequence of channels as selected in these registers is converted. When the sequence bits as shown in [Table 3-9, page 39](#) are set so as to enable the automatic channel sequencer in single pass mode, the sequence starts. The settings in sequencer registers 48h to 4Fh are used to operate the sequence in a user-defined mode of operation. All channels listed in [Table 4-1](#) and [Table 4-2](#) are available to be used in a sequence. For an explanation of the sequencer registers, see [Automatic](#)

Channel Sequencer . When the single pass is complete, the XADC defaults to Single Channel Mode described at the start of this chapter. Thus the XADC converts the channel selected by bits CH5 to CH0 in Configuration Register 0. Another single pass can be started by writing to the sequence bits again. For example, changing to single channel mode (SEQ = 0x3) and back to single pass mode (SEQ = 0x1) restarts the single pass mode and another sequence begins.

The continuous sequence mode is similar to single pass mode; however, the sequence automatically restarts as long as the mode is enabled.

The channel sequencer registers can also be reconfigured through the DRP at run time. The sequencer must first be disabled by writing to sequence bits SEQ3 to SEQ0 before writing to any of the sequencer channel registers. It is recommended that the XADC is placed in default mode by writing zeros to SEQ0 and SEQ1 while updating these registers. The XADC is automatically reset whenever SEQ3 to SEQ0 are written to. The current status register contents are not reset at this time. Restarting the sequencer by writing to bits SEQ3 to SEQ0 resets all channel averaging.

When placed in simultaneous sampling mode, the sequencer automatically sequences through eight pairs of auxiliary analog input channels for simultaneous sampling and conversion as shown in [Table 4-4](#) . This is useful in applications where it is necessary to preserve the phase relationship between two signals.

Auxiliary analog channels 0 to 7 are assigned to ADC A and are nominated as *A channels* . Auxiliary analog channels 8 to 15 are assigned to ADC B and are nominated as *B channels* (see [Figure 1-1, page 8](#)). One A channel and one B channel are always sampled and converted at the same time in simultaneous sampling mode. [Table 4-4](#) shows how A and B pairs are selected using sequencer channel register 49h . Other sequencer registers that define averaging, analog input mode, and settling time are also available for use in this mode.

Channel averaging (4Ah and 4Bh) and analog-input modes (4Ch and 4Dh) can be set on a per channel basis in simultaneous sampling mode. Thus, it is possible to have an A channel configured as unipolar and a B channel configured as bipolar. Bit definitions as shown in [Table 4-1](#) and [Table 4-2](#) apply. The settling time is applied to a channel pair only; thus, setting bit 0 to 1 in register 4Eh sets the settling time to 10 ADCCLKs for both auxiliary channel 0 and channel 8.

Table: Sequencer Register (49h) Bit Definitions for Simultaneous Sampling Mode

Sequence Number	Bit	ADC Channel	Description
1	0	16, 24	Auxiliary channels 0 and 8
2	1	17, 25	Auxiliary channels 1 and 9
3	2	18, 26	Auxiliary channels 2 and 10
4	3	19, 27	Auxiliary channels 3 and 11
5	4	20, 28	Auxiliary channels 4 and 12
6	5	21, 29	Auxiliary channels 5 and 13 ⁽¹⁾
7	6	22, 30	Auxiliary channels 6 and 14 ⁽¹⁾
8	7	23, 31	Auxiliary channels 7 and 15 ⁽¹⁾
x	8	x	Undefined
x	9	x	Undefined
x	10	x	Undefined
x	11	x	Undefined
x	12	x	Undefined

x	13	x	Undefined
x	14	x	Undefined
x	15	x	Undefined

Notes:

1. These simultaneous sampling channels are not supported in Kintex-7 devices. Some other auxiliary analog channels might also not be supported in certain Virtex-7, Artix-7, Spartan-7, and Zynq-7000 SoC device package options. Users should consult the package file for the device.

The on-chip temperature and power supplies can also be included in the sequence of channels monitored by setting the appropriate bits in sequencer register 48h . The second ADC (ADC B) does not carry out a conversion in simultaneous sampling mode when an internal sensor channel is being converted.

Automatic calibration cannot be enabled in simultaneous sampling mode. The XADC must be placed in default mode or other sequencer mode to carry out a calibration. The ADCs are automatically calibrated on power-up, which is sufficient for most applications.

The timing for simultaneous sampling mode is identical to other XADC modes (see [Chapter 5, XADC Timing](#)). Because the ADCs run in lock step, both status registers are updated at exactly the same time. Both continuous and event driven timing modes can be used.

In the independent ADC mode, ADC A is used to implement a fixed “monitoring mode” similar to the default mode except the alarm functions are enabled (see [Automatic Alarms, page 56](#)). In this mode, the alarm outputs are active, and you must correctly configure the alarm threshold. Like default sequencer mode, the averaging is fixed at 16 samples.

ADC B is available to be used with the external analog input channels only. This mode frees up the second ADC for use in a customer application where only a single ADC is required while maintaining monitoring of the FPGA for reliability, safety, and anti-tamper applications.

Only the dedicated channel and auxiliary analog input channels can be assigned to ADC B in this sequencer mode. The internal channels (sensors) are automatically assigned to ADC A, which automatically monitors these channels and generates alarms based on the user-defined alarm thresholds.

As with simultaneous sampling mode, it is not possible to select an automatic calibration of ADC B in this sequencer mode. ADC A is automatically calibrated in this mode. To carry out a calibration on ADC B and maintain on-chip monitoring, the default sequence mode should be selected. The XADC can then be returned to Independent ADC mode after EOS goes High at least once.

Channel selection for independent ADC mode is defined using sequencer channel registers 48h and 49h (see [Table 4-5](#) and [Table 4-6](#)). Internal sensor channels are automatically monitored and calibrated and cannot be assigned to the sequence for ADC B by users. Other sequencer registers that define settling time, analog input mode, and averaging remain unchanged.

Table: Independent ADC Sequencer Mode Bit Definitions (48h)

Sequence Number	Bit	ADC Channel	Description
-	0	-	Not Defined
	1		Not Defined
	2		Not Defined
	3		Not Defined
	4		Not Defined

	5		Not Defined
	6		Not Defined
	7		Not Defined
	8		Not Defined
	9		Not Defined
	10		Not Defined
1	11	3	V_P/V_N
--	12	-	Not Defined
	13		Not Defined
	14		Not Defined
	15		Not Defined

Table: Independent ADC Sequencer Mode Bit Definitions (49h)

Sequence Number	Bit	ADC Channel	Description
2	0	16	Auxiliary channel 0
3	1	17	Auxiliary channel 1
4	2	18	Auxiliary channel 2
5	3	19	Auxiliary channel 3
6	4	20	Auxiliary channel 4
7	5	21	Auxiliary channel 5
8	6	22	Auxiliary channel 6 ⁽¹⁾
9	7	23	Auxiliary channel 7 ⁽¹⁾
10	8	24	Auxiliary channel 8
11	9	25	Auxiliary channel 9
12	10	26	Auxiliary channel 10
13	11	27	Auxiliary channel 11
14	12	28	Auxiliary channel 12
15	13	29	Auxiliary channel 13 ⁽¹⁾
16	14	30	Auxiliary channel 14 ⁽¹⁾
17	15	31	Auxiliary channel 15 ⁽¹⁾

Notes:

1. Auxiliary channels 6, 7, 13, 14, and 15 are not supported on Kintex-7 devices. Some auxiliary analog channels might also not be supported in certain Virtex-7, Artix-7, Spartan-7, and Zynq-7000 SoC device package options. Users should consult the package file for the device.

The timing in this mode of operation is also the same for both ADCs. ADC A monitors the internal sensors at the same sample rate and mode selected for ADC B. Again, ADC B can be operated in continuous or event mode timing. In the case of event mode timing, internal sensor monitoring also stops if the CONVST signal is stopped. Therefore, care should be taken to ensure periodic pulsing of CONVST even when ADC B is not being used by the application, if constant coverage of the on-chip environment is required.

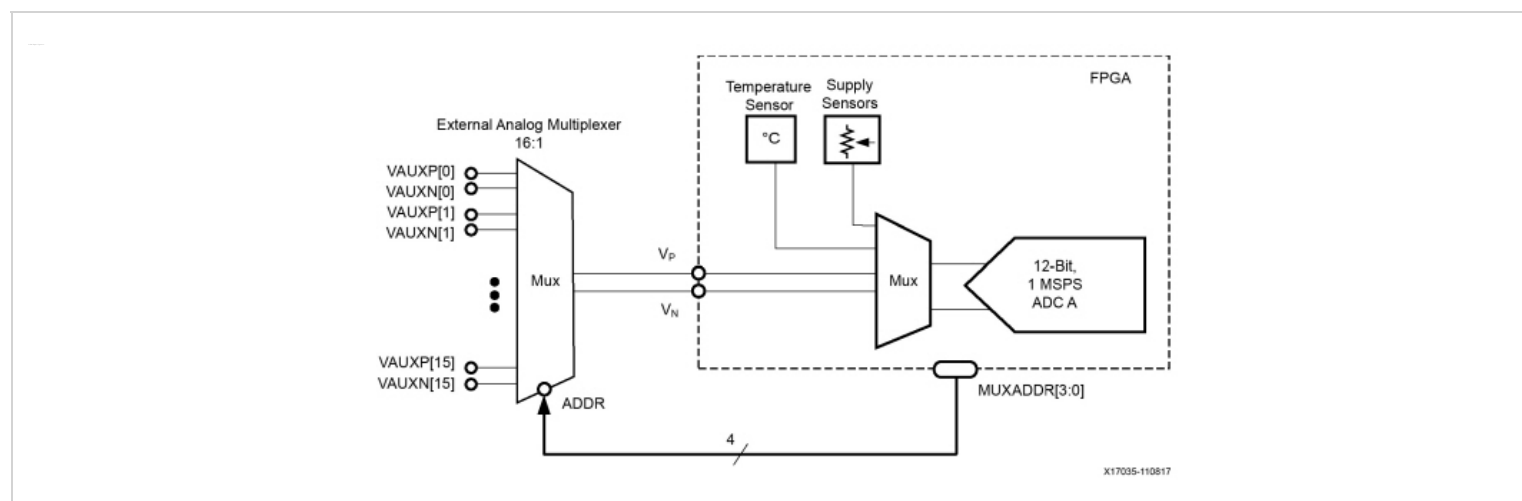
Timing for continuous and event modes remain the same as shown in [Chapter 5, XADC Timing](#).

External Multiplexer Mode

The XADC supports the use of an external analog multiplexer to implement several external analog inputs in situations where FPGA I/O resources are limited and auxiliary analog inputs are not available.

The XADC track/hold amplifiers return to track mode as soon as a conversion starts (see [Figure 4-3](#)). Therefore, the acquisition on the next channel can start during the current conversion cycle. An output bus called MUXADDR[4:0] allows the XADC to control an external multiplexer. The address on this bus reflects the channel currently being acquired, and it changes state as soon as the XADC enters acquisition mode. Users can also nominate the channel to be used with an external multiplexer.

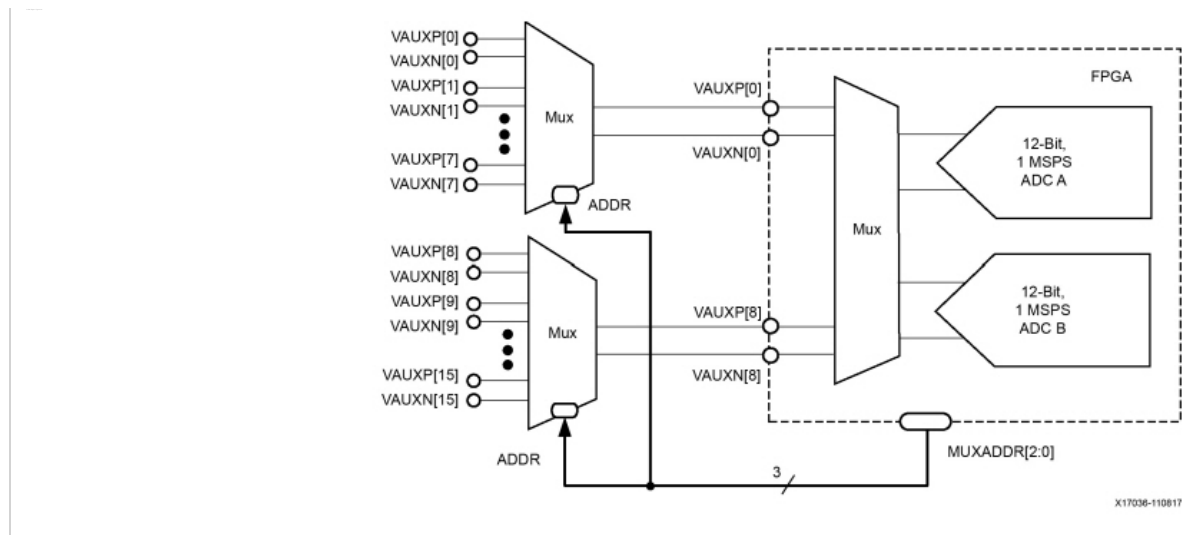
[Figure 4-1](#) illustrates the external multiplexer concept. In this example an external 16:1 analog multiplexer is used instead of consuming the 32 FPGA I/Os required to implement the 16 auxiliary analog input channels using the internal multiplexer. Any four FPGA I/Os can be used for the external multiplexer decode operation. As shown in [Figure 4-1](#), the dedicated analog inputs (V_P , V_N) are used to connect the external multiplexer to the XADC block, thereby making 16 analog inputs available. The external multiplexer mode of operation is enabled by setting the MUX bit in Configuration Register 0 (see [Control Registers, page 35](#)).



External Multiplexer Mode

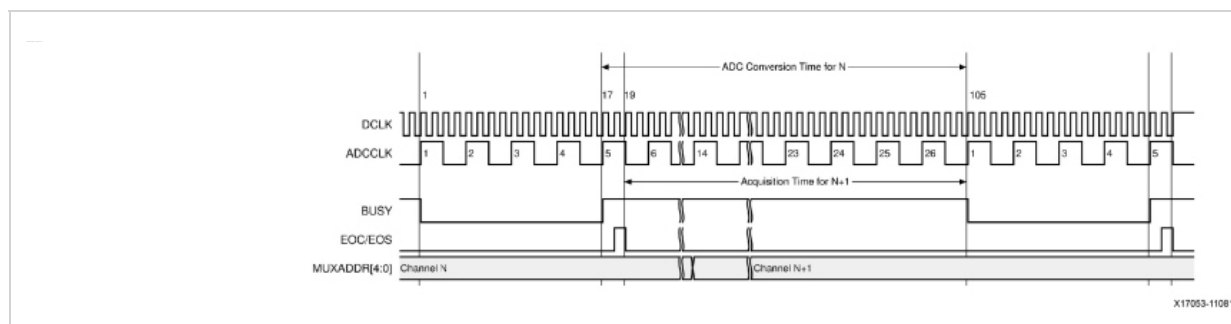
When the MUX bit is set to a 1, the channel selection bits (CH0 to CH4) in Configuration Register 0 are used to nominate the channel for connection to the external multiplexer. For example, as shown in [Figure 4-1](#), the dedicated analog input channel V_P / V_N is used. In this case, channel 3 (00011b) should be written to CH4 to CH0 in Control Register 40h. Any one of the auxiliary channels can also be used for connection to the external multiplexer. In the case of simultaneous sampling mode (see [Simultaneous Sampling Mode, page 50](#)), two channels must be allocated to two external multiplexers to support simultaneous sampling.

[Figure 4-2](#) illustrates how the external multiplexer mode is implemented for simultaneous sampling mode. The channels selected for connection are also selected by writing to CH4 to CH0 but are allocated in pairs as defined in [Table 4-4](#). For example, writing 16 (10000) to CH4 to CH0 would select auxiliary channels 0 and 8 for connection to external multiplexers as shown in [Figure 4-2](#).



External Multiplexer Mode for Simultaneous Sampling

In both cases, the MUXADDR[4:0] bus is used to automatically select the external multiplexer channel. [Figure 4-3](#) shows how the MUXADDR bus reflects the next channel selection ($N + 1$). MUXADDR changes state eight ADCCLK cycles after BUSY goes High.



External Multiplexer Timing (Continuous Sampling Mode)

Maximum and Minimum Status Registers

The XADC also tracks the minimum and maximum values recorded for the internal sensors since the last power-up or since the last reset of the XADC control logic. The maximum and minimum values recorded are stored in the DRP status registers starting at address 20h (see [Status Registers, page 30](#)). On power-up or after reset, all minimum registers are set to FFFFh and all maximum registers are set to 0000h. Each new measurement generated for an on-chip sensor is compared to the contents of its maximum and minimum registers. If the measured value is greater than the contents of its maximum registers, the measured value is written to the maximum register. Similarly, for the minimum register, if the measured value is less than the contents of its minimum register, the measured value is written to the minimum register. This check is carried out every time a measurement result is written to the status registers.

Table: Maximum and Minimum Registers

Status Register	Description
20h	Temp max – Maximum temperature recorded
21h	V _{CCINT} max – Maximum V _{CCINT} recorded
22h	V _{CCAUX} max – Maximum V _{CCAUX} recorded
23h	V _{CCBRAM} max – Maximum V _{CCBRAM} recorded

24h	Temp min – Minimum temperature recorded
25h	V_{CCINT} min – Minimum V_{CCINT} recorded
26h	V_{CCAUX} min – Minimum V_{CCAUX} recorded
27h	V_{CCBRAM} min – Minimum V_{CCBRAM} recorded
28h	$V_{CCPINT}^{(1)}$ max – Maximum V_{CCPINT} recorded
29h	$V_{CCPAUX}^{(1)}$ max – Maximum V_{CCPAUX} recorded
2Ah	$V_{CCO_DDR}^{(1)}$ max – Maximum V_{CCO_DDR} recorded
2Bh	Unassigned
2Ch	$V_{CCPINT}^{(1)}$ min – Minimum V_{CCPINT} recorded
2Dh	$V_{CCPAUX}^{(1)}$ min – Minimum V_{CCPAUX} recorded
2Eh	$V_{CCO_DDR}^{(1)}$ min – Minimum V_{CCO_DDR} recorded
2Fh	Unassigned

Notes:

1. Only available on the Zynq-7000 SoC devices.

Automatic Alarms

The XADC also generates an alarm signal on the logic outputs ALM[7:0] when an internal sensor measurement (Temperature, V_{CCINT} , V_{CCAUX} , V_{CCBRAM} , V_{CCPINT} , V_{CCPAUX} , or V_{CCO_DDR}) exceeds some user-defined thresholds. Only the values written to the status registers are used to generate alarms. If averaging has been enabled for a sensor channel, the averaged value is compared to the alarm threshold register contents. The alarm outputs are disabled by writing a 1 to bits ALM6 to ALM0 in configuration register 1. The alarm thresholds are stored in control registers 50h to 5Fh. [Table 4-8](#) defines the alarm thresholds that are associated with specific control registers. The limits written to the threshold registers are MSB justified. Limits are derived from the temperature and power-supply sensor transfer functions (see [Figure 2-9, page 26](#) and [Figure 2-10, page 27](#)).

Table: Alarm Threshold Registers

Control Register	Description	Alarm
50h	Temperature upper	ALM[0]
51h	V_{CCINT} upper	ALM[1]
52h	V_{CCAUX} upper	ALM[2]
53h	OT alarm limit (1)	OT
54h	Temperature lower	ALM[0]
55h	V_{CCINT} lower	ALM[1]
56h	V_{CCAUX} lower	ALM[2]
57h	OT alarm reset (1)	OT

58h	V _{CCBRAM} upper	ALM[3]
59h	V _{CCPINT} upper ⁽²⁾	ALM[4]
5Ah	V _{CCPAUX} upper ⁽²⁾	ALM[5]
5Bh	V _{CCO_DDR} upper ⁽²⁾	ALM[6]
5Ch	V _{CCBRAM} lower	ALM[3]
5Dh	V _{CCPINT} lower ⁽²⁾	ALM[4]
5Eh	V _{CCPAUX} lower ⁽²⁾	ALM[5]
5Fh	V _{CCO_DDR} lower ⁽²⁾	ALM[6]

Notes:

1. OT alarm limit and OT alarm reset are described in [Thermal Management](#) .
2. Only Zynq-7000 SoC devices support these channels.

When the measured value on the supply sensor for V_{CCINT}, V_{CCAUX}, V_{CCBRAM}, V_{CCPINT}, V_{CCPAUX}, or V_{CCO_DDR} is greater than the thresholds in control registers 51h, 52h, 58h, 59h, 5Ah, and 5Bh or less than the thresholds in control registers 55h, 56h, 5Ch, 5Bh, 5D, and 5Eh, respectively, the output alarms go active. The alarms are reset when a subsequently measured value falls inside the threshold.

The on-chip temperature measurement is used for critical temperature warnings and also supports automatic shutdown to help prevent the device from being permanently damaged. The on-chip temperature measurements record the junction temperatures continuously during pre-configuration and automatic shutdown.

Automatic shutdown must be enabled by setting the four LSBs (DI[3:0]) to 0011b in the OT upper alarm register (53h) and by adding the following constraint to the project XDC file:

```
set_property BITSTREAM.CONFIG.OVERTEMPPowerDOWN ENABLE [current_design]
```

The default over temperature (OT) threshold is 125°C. The 125°C threshold is used when the contents of the OT upper alarm register (53h) is 0000h, including pre-configuration. To override this default condition, the 12 MSBs of the OT upper register (control register 53h) must be set using the temperature sensor transfer function (see [Figure 2-10](#)).

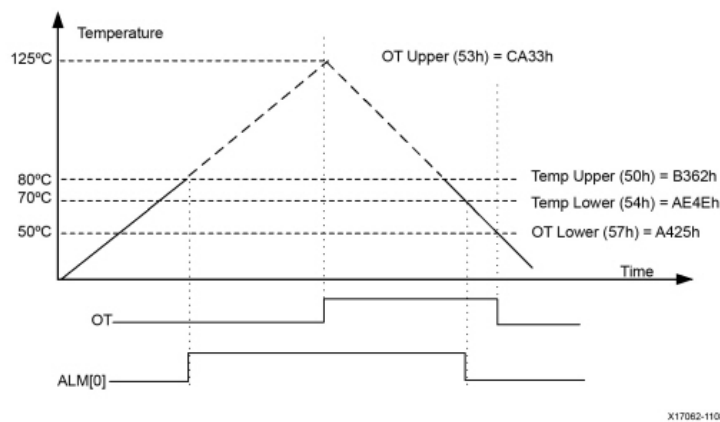
$$12\text{-bit OT Alarm Limit} = (Temp + 273.15) \times (4096 / 503.975)$$

$$OT\ Temp(^{\circ}C) = \frac{(12\text{ bit ADC Code}) \times 503.975}{4096} - 273.15$$

From Equation [Equation 4-1](#) (in bits) or Equation [Equation 4-2](#) for 125°C, control register 53h must be set to CA3h for the 12 MSBs. Because the four LSBs must be set to 3h, this gives the 16-bit register value of CA33h for control register 53h .

$$Temp\ upper/lower (^{\circ}C) = \frac{(16\text{ bit ADC Code}) \times 503.975}{65536} - 273.15$$

For the remaining temperature thresholds, use [Equation 4-3](#) to define the 16-bit ADC code values.



Thermal Management Operation Example

As shown in Figure 4-4, when the die temperature exceeds the OT upper threshold (or the default 125°C), the over-temperature alarm logic output becomes active and 10 ms later the device initiates the shutdown sequence. When the automatic shutdown starts, the device is disabled and GHIGH is asserted to prevent any contention (see UG470, *7 Series FPGAs Configuration User Guide* [Ref 3]). When OT is deasserted (50°C as shown in Figure 4-4), GHIGH is also deasserted and the start-up sequence is initiated releasing all global resources.

Note: When the OT alarm has been triggered in 7 series FPGAs, PROGRAM_B input levels are ignored until the die temperature resets the OT alarm (OT_Lower 57h). That is, PROGRAM_B can only be used to reconfigure the FPGA after GHIGH has been deasserted.

While the device is shut down, XADC automatically uses the internal clock oscillator, but otherwise remains unchanged. JTAG is only guaranteed to 125°C.

The automatic shutdown feature is for preventing permanent damage to the device. After the temperature has fallen below the OT lower (57h) setting and OT is deasserted, the device must be reconfigured. Additionally, because a catastrophic failure occurred, all power to the device should be removed and it should be determined why the device temperature increased so dramatically. Designs should use a thermal management procedure with the temperature alarm (ALM[0]) to actively control the device temperature during operation.

If XADC is instantiated, the automatic shutdown feature can be disabled either by setting the OT signal within Config Reg 1 (41h) High, or by adding the following constraint to the project XDC file:

```
set_property BITSTREAM.CONFIG.OVERTEMPPowerDOWN DISABLE [current_design]
```

See UG908, *Vivado Design Suite User Guide: Programming and Debugging* [Ref 8] for additional details on device configuration bitstream settings. ENABLE and DISABLE are allowable values.

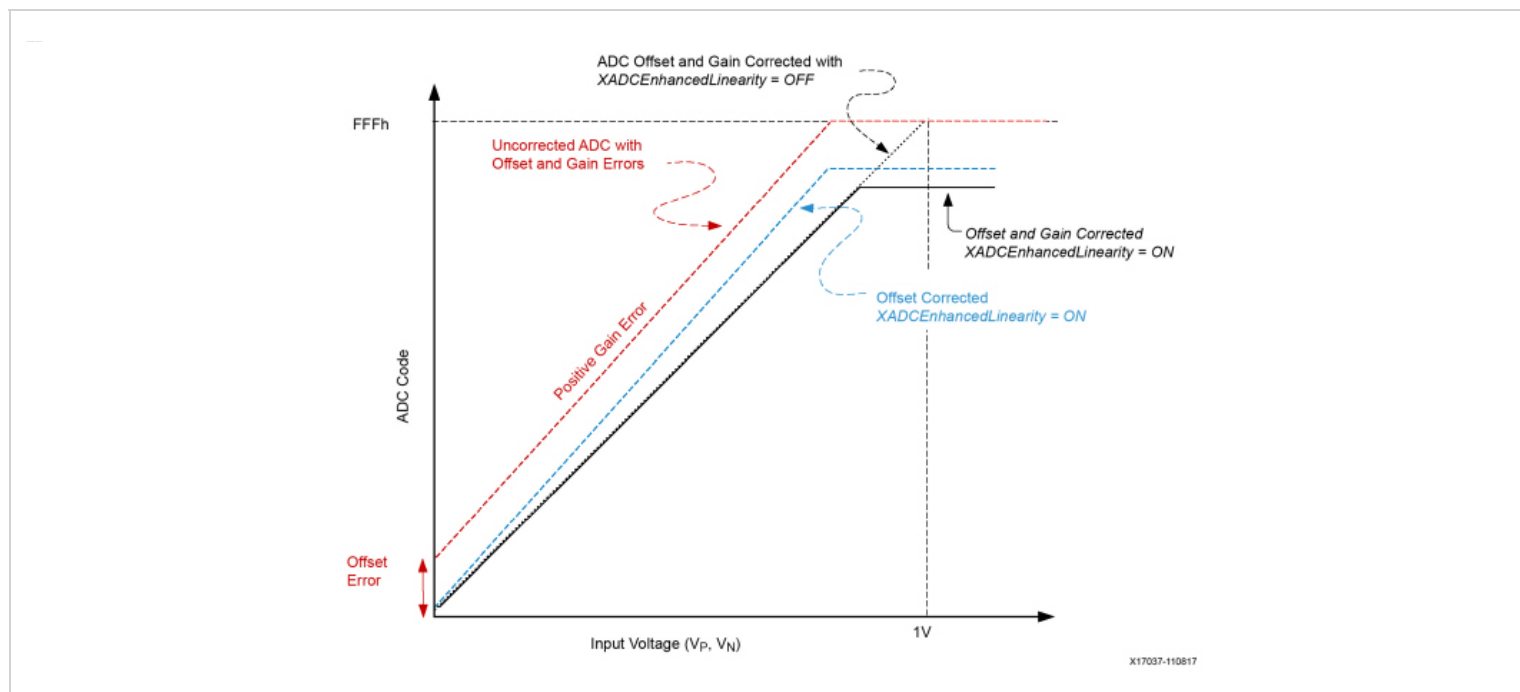
A second user-programmable temperature threshold level (temperature upper 50h) is used to carry out a user-defined thermal management procedure, such as powering on or controlling the speed of a fan. Alarm signal ALM[0] is High when the device temperature exceeds the limit in the temperature upper control register 50h. ALM[0] remains High until the temperature falls below the lower threshold, temperature lower (54h). As shown in Figure 4-4, this means that ALM[0] is High when the temperature reaches 80°C and remains High until the temperature falls to 70°C. This operation differs for the supply sensor alarm because the supply alarm resets when the measurement is between the upper and lower thresholds.

For ISE design tools, a BitGen option called *XADCEnhancedLinearity* is available. It enhances the linearity (INL) and dynamic performance (SNR and THD) of the ADCs in the XADC block. This bitgen -g command line option is *XADCEnhancedLinearity = [ON|OFF]*. By default this BitGen option is OFF. This mode was also OFF by default in all previous releases of the ISE design tools.

For the Vivado design tools, use *XADCENHANCEDLINEARITY*. See UG908, *Vivado Design Suite User Guide: Programming and Debugging* [Ref 8] for the XDC templates.

```
set_property BITSTREAM.GENERAL.XADCENHANCEDLINEARITY <On|Off> [current_design]
```

If the *XADCEnhancedLinearity* option is not enabled (that is, OFF), there is no change to the XADC operation when offset and gain calibration is enabled. See [XADC Calibration Coefficients, page 33](#), for an explanation of how the ADC offset and gain calibration is enabled. However, if this new BitGen option is enabled, some differences in the ADC transfer functions are observed when offset and gain calibration coefficients are applied.



ADC Calibration with *XADCEnhancedLinearity* = ON

Figure 4-5 illustrates the impact of enabling offset and gain error correction with *XADCEnhancedLinearity* = ON. In this example, the red line shows the impact of offset and gain errors on the ADC transfer function. This is exaggerated for clarity. To remove the offset error, a constant correction (offset calibration coefficient) is subtracted from the ADC output. This is shown as the blue line in Figure 4-5. The black solid line shows the impact of the gain error and offset correction. Because the ADC uncalibrated output reached full scale (FFFh) before the input reached 1V (due to offset and gain errors), part of the ADC transfer function is no longer available even after calibration. Thus, the ADC maximum code stays at a value less than FFFh even for input voltages of 1V. This clipping range is typically no more than about 10 mV around either 0V or 1V input, depending on the ADC offset and gain errors. For example, if the offset error was negative in the example shown in Figure 4-5, ADC would produce a 0 code out until the analog input exceeded this offset with offset correction enabled.

Note: This clipping effect is not seen when the option *XADCEnhancedLinearity* = OFF.

XADC Timing

All XADC timing is synchronized to the DRP clock (DCLK). The ADCCLK is generated by dividing DCLK by the user selection in configuration register 2 (see [Control Registers, page 35](#)). ADCCLK is an internal clock used by the ADCs and is not available externally. ADCCLK is only included here to aid in describing the timing.

The ADC block is operated in one of two possible timing modes, continuous sampling mode and event driven sampling mode.

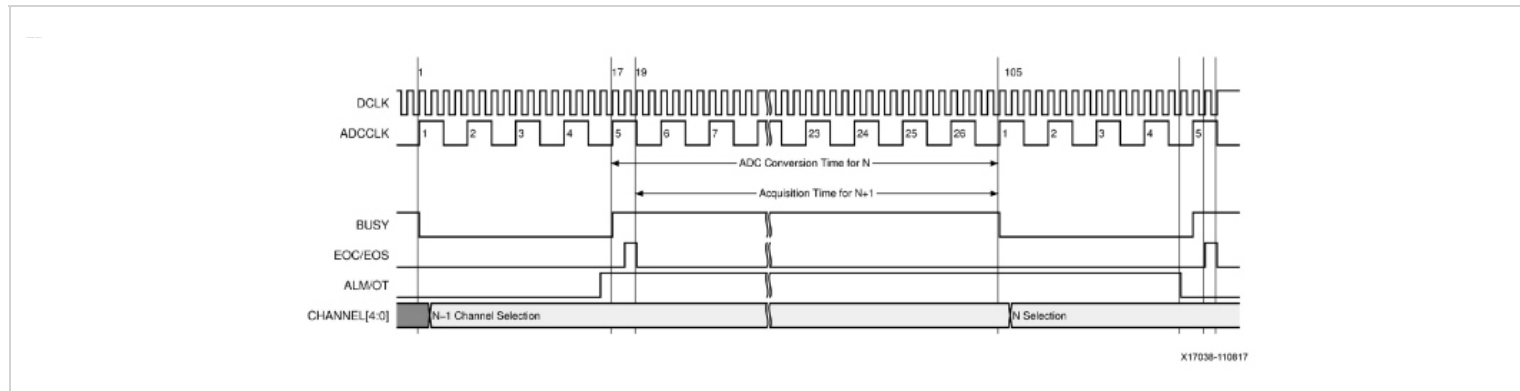
In continuous sampling mode, the ADC automatically starts a new conversion at the end of a current conversion cycle. In event sampling mode, you must initiate the next conversion after the current conversion cycle ends by using the CONVST or CONVSTCLK inputs. The operating mode is selected by writing to configuration register 0 (see [Control Registers](#)).

Refer to the respective 7 series FPGAs data sheet for the latest XADC timing specifications. The robust nature of the XADC ensures continued and correct operation even if the external clock input DCLK is stopped. In this situation, the XADC automatically switches over to an internal clock oscillator to continue to operate as configured. After configuration, the DCLK

input requires 20 DCLKs to resynchronize to the external clock. The JTAGLOCKED signal is High during this period. It is not possible to access the DRP until resynchronization has occurred.

Continuous Sampling

In continuous sampling mode, the ADCs continue to carry out a conversion on the selected analog inputs. [Figure 5-1](#) shows the timing associated with continuous sampling mode. The ADCCLK is generated by a clock divider. The analog-to-digital conversion process is made up of two parts, the acquisition phase and the conversion phase.



Continuous Sampling Mode

During the acquisition phase, the ADC acquires the voltage on a selected channel to perform the conversion. The acquisition phase involves charging a capacitor in the ADC to the voltage on the selected channel (see [Analog Input Description, page 23](#) for more information). The time required to charge this capacitor depends on the selected input-channel source impedance. The XADC allows the acquisition of the next channel to start during the current conversion. This is possible because the ADC has a separate track-and-hold amplifier (T/H). Thus, when the ADC starts to convert an input voltage, the T/H is free to start precharging to the next voltage to be converted. This allows for a faster sampling rate because the ADC does not need to acquire the full input voltage at the end of the current conversion. By default, a *settling period* of four ADCCLK cycles is left between the end of the current conversion and the start of the next conversion. This can be extended to 10 ADCCLK cycles by setting the ACQ bits in either the control registers (single channel mode) or the sequence registers (sequencer modes). The additional ADCCLK cycles are useful when the external channel has a large source impedance (greater than 10 k Ω).

When operating in single channel mode (see [Chapter 4, XADC Operating Modes](#)), you must write to configuration register 0 to select the next channel for conversion. The address of the next channel to convert is read from the control registers at the end of the previous conversion (when BUSY goes Low), and the analog signal is acquired during the current conversion. Additional acquisition or settling time after the end of the current conversion is configured by the ACQ bit in configuration register 0.

For more information on the effects of source impedance on the acquisition, see [Analog Input Description](#).

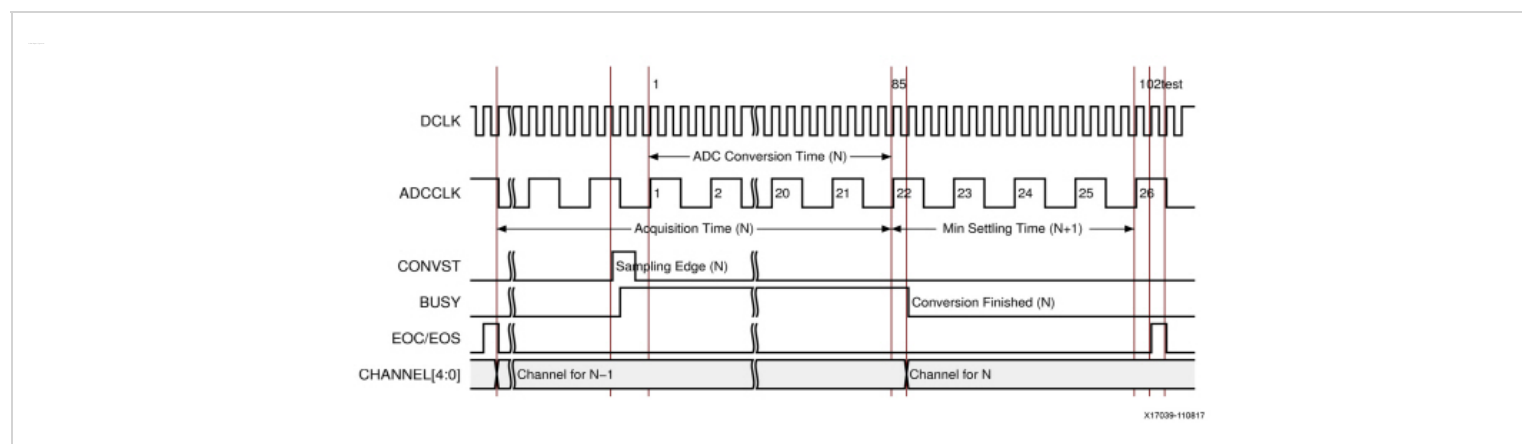
The conversion phase starts on the sampling edge (next rising edge of DCLK) at the end of the 4 or 10 ADCCLK cycles settling time. The BUSY signal transitions to an active-High on the next rising edge of DCLK to indicate the ADC is carrying out a conversion. The conversion phase is 22 ADCCLK cycles long. 16 DCLK cycles after BUSY goes Low, EOC pulses High for one DCLK cycle when the conversion results have been transferred to the output registers. EOS indicates the end of a sequence, which depends on the automatic channel sequencer settings and averaging settings. If the automatic channel sequencer is used, then EOS matches the last channel enabled (see [Table 4-1](#)). When averaging is used, EOS only pulses High after all the samples have been completed (16, 64, and 256). The number of samples is set by AVG0 and AVG1 in Configuration Reg #0 (40h) (see [Table 3-8](#)).

When XADC is being operated in a sequence mode, you can identify the channel being converted by monitoring the channel address (CHANNEL[4:0]) logic outputs. The multiplexer channel address of the channel being converted is updated on these logic outputs when BUSY transitions Low at the end of the conversion phase. The channel address outputs can be used with the EOC and DRDY signals to automatically latch the contents of the output data registers into a FIFO or block RAM. This is accomplished by connecting the CHANNEL[4:0] outputs to DADDR[4:0] (with DADDR[6:5] = 0), using EOC as a DEN (enable) for the DRP, and using DRDY as a WE (write enable) for the block RAM.

XADC EOS signal has the same timing as EOC. This signal is pulsed when the output data register for the last channel in a programmed channel sequence is updated.

Event-Driven Sampling

Figure 5-2 illustrates the event-driven sampling mode. In this operating mode, the sampling instant and subsequent conversion process are initiated by a trigger signal called convert start (CONVST). Event-driven sampling mode is used when precise control over the sampling instant is necessary.



Event Driven Sampling Mode

The DCLK must always be present when using event-driven sampling mode. If no DCLK is present, the XADC reverts to continuous mode timing using an internal clock oscillator.

A Low-to-High transition (rising edge) on CONVST or CONVSTCLK defines the exact sampling instant for the selected analog input channel. The BUSY signal transitions High just after the sampling instant on the next rising edge of DCLK. CONVST can be an asynchronous external signal in which case the XADC automatically resynchronizes the conversion process to the ADCCLK.

As with the continuous sampling mode, enough time must be allowed for the acquisition phase, that is, the time between a channel change and the sampling edge (the rising edge of CONVST or CONVSTCLK, see [Analog Input Description, page 23](#)). This allows the ADC to acquire the new signal before it is sampled by the CONVST signal and the conversion phase starts. A settling period of four ADCCLK cycles is recommended between the end of the current conversion (BUSY going Low) and the start of the next conversion. The ACQ bit has no meaning in event-sampling mode because the sampling instant is controlled by CONVST/CONVSTCLK. Therefore, the acquisition time on a selected channel is also controlled by CONVST/CONVSTCLK. CONVST and CONVSTCLK are logically ORed within the XADC. The T/H starts to acquire the voltage on the next channel as soon as BUSY goes High and a conversion starts.

After the analog input has been sampled by a rising edge on CONVST/CONVSTCLK, a conversion is initiated on the next rising edge of ADCCLK. After a conversion has been initiated by CONVST, it is not possible to interrupt the conversion or start a new conversion until BUSY goes Low.

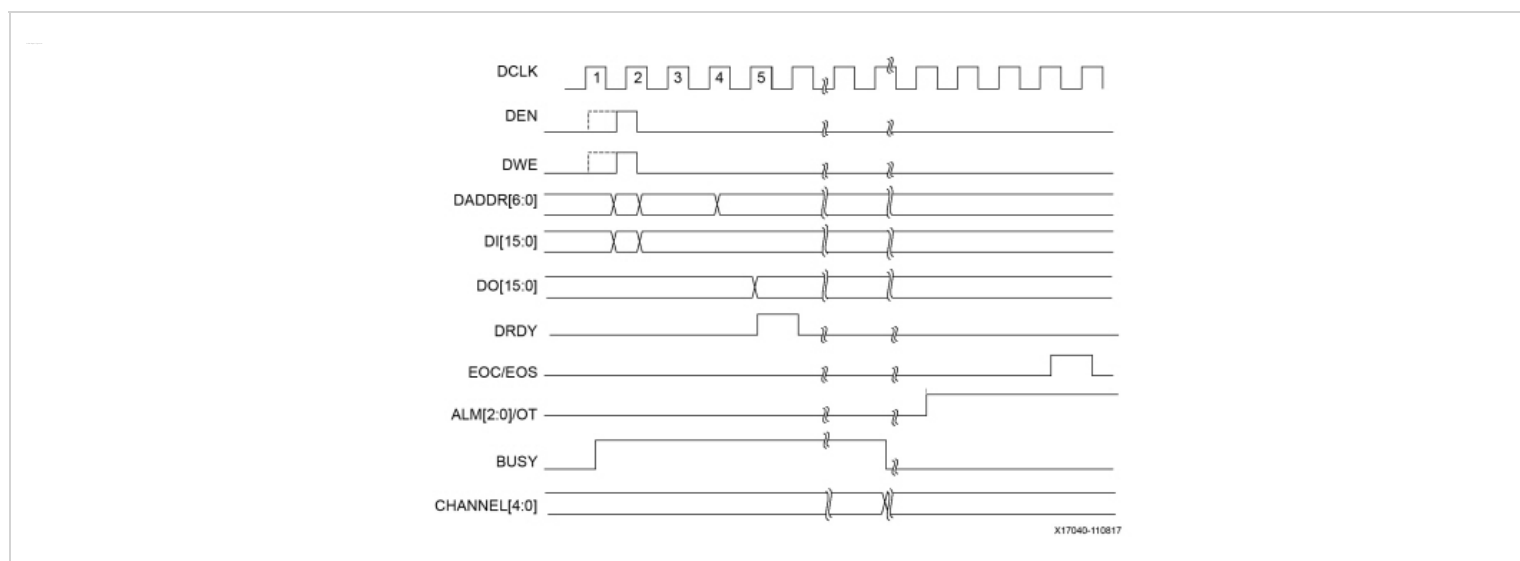
16 DCLK cycles after BUSY goes Low, EOC pulses High for one DCLK cycle when the conversion result has been transferred to the output register. EOS indicates the end of a sequence that depends on the automatic channel sequencer settings and averaging settings. If the automatic channel sequencer is used, then EOS matches the last channel enabled (see [Table 4-1](#)). When averaging is used, EOS only pulses High after all the sequences or samples have been completed (16, 64, and 256). The number of samples is set by AVG0 and AVG1 in Configuration Reg #0 (40h) (see [Table 3-8](#)).

Remember that CONVST/CONVSTCLK starts a single conversion. When using the automatic channel sequencer or averaging, the number of conversions are the product of the number of channels in a sequence and the number of samples being averaged.

Dynamic Reconfiguration Port (DRP) Timing

Figure 5-3 illustrates a DRP read and write operation. When the DEN is logic High, the DRP address (DADDR) and write enable (DWE) inputs are captured on the next rising edge of DCLK. DEN should only go high for one DCLK period.

If DWE is logic Low, a DRP read operation is carried out. The data for this read operation is valid on the DO bus when DRDY goes high. Thus DRDY should be used to capture the DO bus. For a write operation, the DWE signal is logic High and the DI bus and DRP address (DADDR) is captured on the next rising edge of DCLK. The DRDY signal goes logic High when the data has been successfully written to the DRP register. A new read or write operation cannot be initiated until the DRDY signal has gone low.



DRP Detailed Timing

Application Guidelines

The 7 series FPGAs XADC is a precision analog measurement system based on a 12-bit analog-to-digital converter (ADC) with an LSB size approximately equal to 250 μ V. To achieve the best possible performance and accuracy with all measurements (both on-chip and external), several dedicated pins for the ADC reference and power supply are provided. Care must be taken with the connection of these pins to ensure the best possible performance from the ADC. This chapter outlines the basic design guidelines to consider as part of the requirements for board design.

Reference Inputs (VREFP and VREFN)

These high-impedance inputs are used to deliver a differential reference voltage for the analog-to-digital conversion process. ADCs are only as accurate as the reference provided. Any reference-voltage error results in a gain error versus the ideal ADC transfer function (see [Chapter 2, Analog-to-Digital Converter](#)). Errors in the reference voltage affect the accuracy of absolute measurements for both on-chip sensors and external channels. Noise on the reference voltage also adds noise to the ADC conversion and results in more code transition noise or poorer than expected SNR.

For typical usage, the reference voltage between V_{REFP} and V_{REFN} should be maintained at $1.25V \pm 0.2\%$ using an external reference IC. Reference voltage ICs that deliver 1.25V are widely available from several vendors. Many vendors offer reference voltage ICs in small packages (SOT-23 and SC70). The 1.25V reference should be placed as close as possible to the reference pins and connected directly to the V_{REFP} input, using the decoupling capacitors recommended in the reference IC data sheet. The recommended reference connections are illustrated in [Figure 6-1](#).

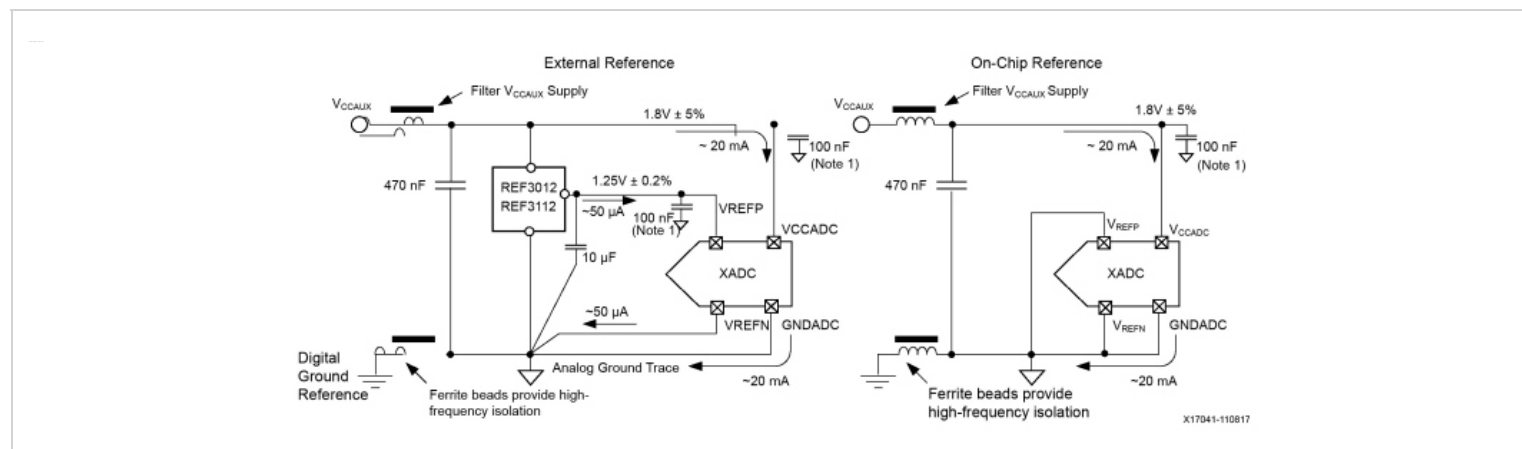
The XADC also has an on-chip reference option which is selected by connecting V_{REFP} and V_{REFN} to ADCGND as shown in [Figure 6-1](#). Due to reduced accuracy, the on-chip reference does impact the measurement performance of the XADC as explained previously.

Analog Power Supply and Ground (VCCADC and GNDADC)

These inputs provide the power supply and ground reference for the analog circuitry in the XADC. A common mechanism for the coupling of noise into an analog circuit is from the power supply and ground connections. Excessive noise on the analog supply or ground reference affects the ADC measurement accuracy. For example, I/O switching activity can cause significant disturbance of the digital ground reference plane. Thus, it would not be advisable to use the digital ground as an analog ground reference for XADC.

Similarly, for the digital supplies for the FPGA logic, high switching rates result in high-frequency voltage variations on the supply, even with decoupling. In an effort to mitigate these effects on the ADC performance, a dedicated supply and ground reference is provided. [Figure 6-1](#) illustrates how to use the 1.8V V_{CCAUX} supply to power the analog circuitry. V_{CCAUX} is filtered using a low-pass network. The filter design depends on the ripple and ripple frequency (if any) on the V_{CCAUX} supply if, for example, a switching regulator is used. There is also a power-supply rejection specification for the external reference circuit which needs to be considered. The filtering should ensure no more than 1 LSB (250 μ V) of noise on the reference output to minimize any impact on ADC accuracy at 12 bits.

The other source of noise coupling into the ADC is from the ground reference GNDADC. In mixed-signal designs, it is common practice to use a separate analog ground plane for analog circuits to isolate the analog and digital ground return paths to the supply. Common ground impedance is a mechanism for noise coupling and needs to be carefully considered when designing the PCB. This is shown in [Figure 2-4, page 21](#), where the common ground impedance R_G converts digital switching currents into a noise voltage for the analog circuitry. While a separate analog ground plane is recommended for 12-bit operation, it is often not possible or practical to implement a separate analog ground plane in a design. For example, if a user only intends to use the on-chip sensors, one low-cost solution is to isolate V_{REFN} and GNDADC ground references (such as a trace) from the digital ground (plane) using a ferrite bead as shown in [Figure 6-1](#). This is illustrated in [PC Board Design Guidelines](#).



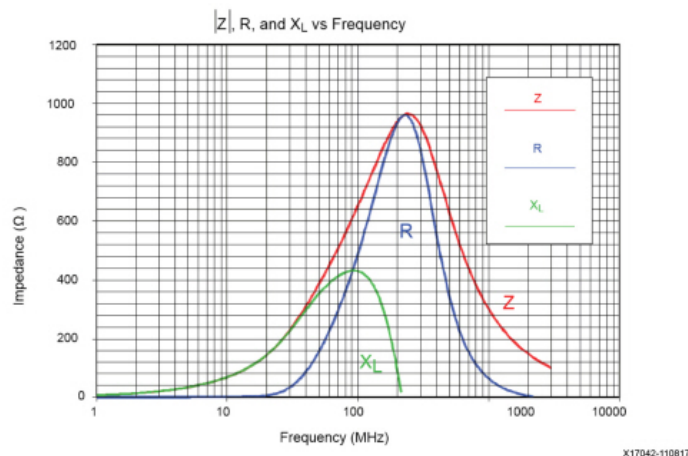
ADC Power and Ground Connections

Notes relevant to [Figure 6-1](#) :

1. Place the 100 nF capacitor as close as possible to the package balls (see [PC Board Design Guidelines, page 68](#)).

The ferrite bead behaves like a resistor at high frequencies and functions as a lossy inductor. A typical ferrite impedance versus frequency plot is shown in [Figure 6-2](#). The ferrite helps provide high frequency isolation between digital and analog grounds. The reference IC maintains a 1.25V difference of between V_{REFP} and V_{REFN} . The ferrite offers little resistance to the analog DC return current.

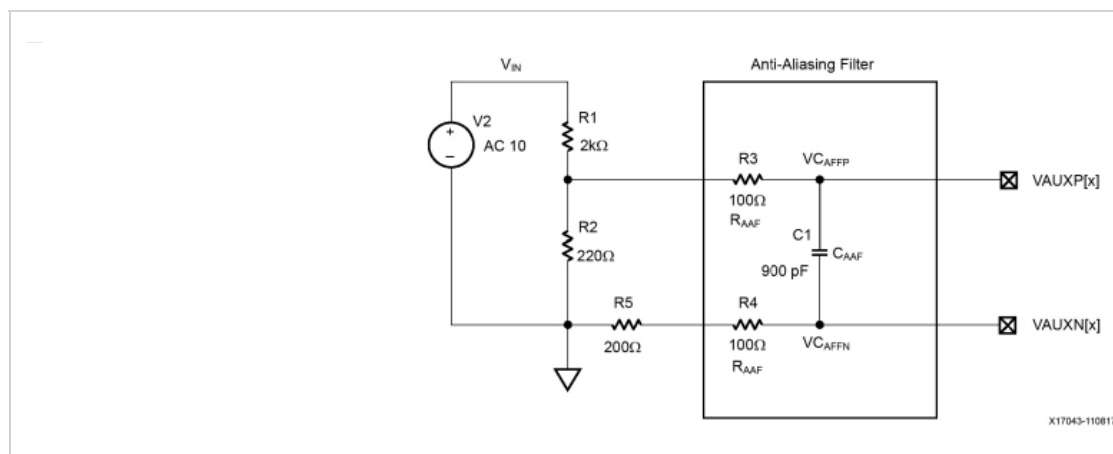
The reference inputs should be routed as a tightly coupled differential pair from the reference IC to the package pins. If routed on the same signal layer, the supply and analog ground traces (V_{CCADC} and GNDADC) should be used to shield the reference inputs because they have a higher tolerance to any coupled noise.



Ferrite Bead Characteristic

External Analog Inputs

The analog inputs are high-impedance differential inputs. The differential input scheme enables the rejection on common mode noise on any externally applied analog-input signal. Because of the high impedance of each input (such as V_P and V_N), the input AC impedance is typically determined by the sensor, the output impedance of the driving circuitry, or other external components. Figure 6-3 illustrates a simple resistor divider network is used to monitor an external 2.5V supply rail in unipolar input mode. Analog-input traces on the PCB should also be routed as tightly coupled differential pairs.



Voltage Attenuation

See XAPP795, *Driving the Xilinx Analog-to-Digital Converter Application Note* [Ref 7] for more details. In Figure 6-3, resistors $R1$ and $R2$ divide a 10V supply down to 1V to work with the XADC. $R5$ has been impedance matched to the parallel resistance of $R1$ and $R2$.

The anti-aliasing filter settling time for this example is determined by Equation 6-1. With a resolution of 12 bits, the example components result in a settling time of 4.9×10^{-6} s or slightly faster than 200 Ks/s.

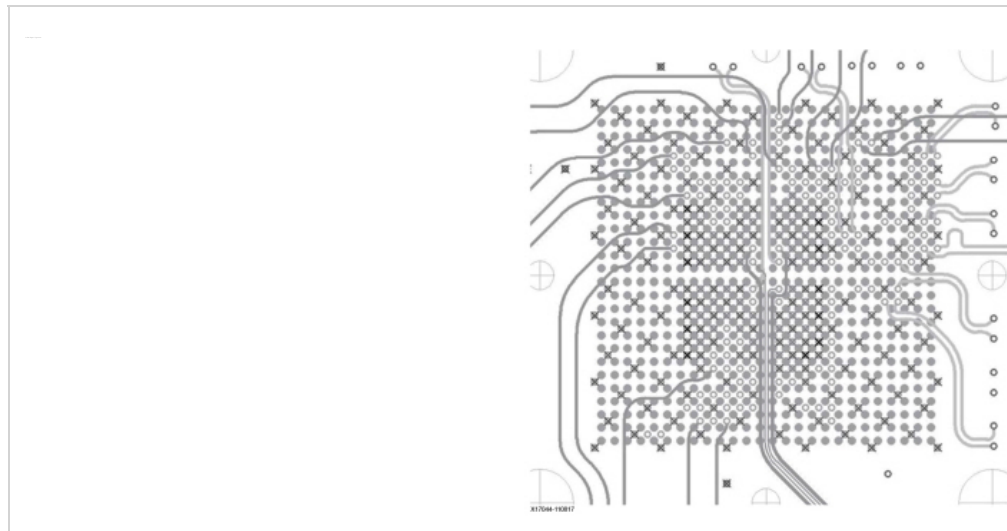
$$T_{\text{settling}} = \ln(2^{\text{Resolution} - 1}) \times \left(\frac{R1 \times R2}{R1 + R2} + R3 + R4 + R5 \right) \times C1 = 4.9 \times 10^{-6} [s]$$

Also shown in Figure 6-3, is a low-pass filter network at the analog differential inputs. This filter network is commonly referred to as the anti-alias filter and should be placed as close as possible to the package pins. The sensor can be placed remotely from the package as long as the differential input traces are closely coupled. The anti-alias filter attenuates high-frequency signal components entering the ADC where they could be sampled and aliased, resulting in ADC measurement corruption. A discussion of aliasing in sampled systems is beyond the scope of this document. A good data converter reference book can provide more information on this topic.

The input voltage can exceed V_{CCADC} (1.8V) or go below $GNDADC$ by as much as 100 mV without damage to the XADC. To limit the current to 1.0 mA, a current-limiting resistor of at least 100 Ω should be placed in series with the analog inputs. The resistors in the anti-alias filters fulfill this requirement. If the analog input range (1V) is exceeded, the ADC output code clips at the maximum output code shown in [Figure 2-2, page 19](#) or [Figure 2-3, page 20](#), depending on the analog input mode. Negative input voltages clip at zero code.

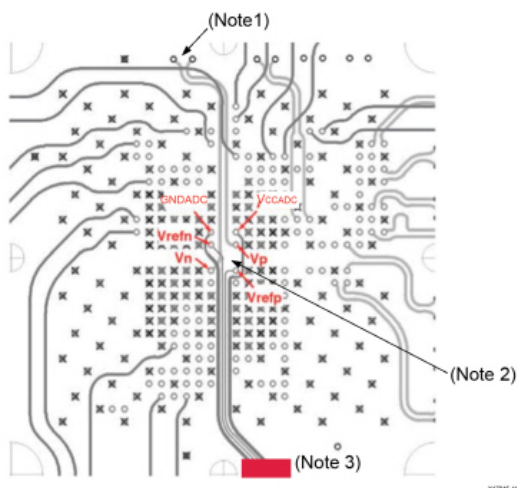
PC Board Design Guidelines

[Figure 6-4](#) and [Figure 6-5](#) illustrate one possible way to address the requirements outlined in the previous sections. [Figure 6-4](#) shows how, by staggering the vias with respect to the pads, north-south and east-west routing channels through the via field are created. These routing channels can be used to bring tightly coupled differential pairs into the center of the via field, even when using 5 mil tolerances.



Routing Channels to Center of Array Created by Staggering Vias

Reference Inputs (V_{REFP} and V_{REFN}) should be Routed as Differential Pairs into the Center of the BGA



Notes relevant to [Figure 6-5](#) :

1. Place anti-alias filters for analog inputs close to the FPGA.
2. Place 100 nF decoupling for V_{REFP} and V_{CCADC} here.
3. Place external reference IC and any dedicated analog power supply regulation as close to the FPGA as possible.

In [Figure 6-5](#), the pads have been removed for clarity. The reference inputs (V_{REFP} and V_{REFN}) are routed as a tightly coupled differential pair from an external 1.25V reference IC at the bottom edge of the FPGA (refer to [Figure 6-1](#) for the connections). The analog power supply and ground reference are also routed into the center of the BGA using traces. No power planes are required to supply a ground reference for the XADC. The analog supply and ground reference are connected to the external reference IC as shown in [Figure 6-1](#). In the PCB implementation shown, the supply and ground traces are routed on either side of the reference traces on the same signal layer and act as guards between the reference traces and any potential aggressors (for example, clocks and switching I/Os). It is not a requirement that the supply and ground traces are routed on the same signal layer as shown, but they should be routed on an adjacent layer. The V_{REFN} and GNDADC traces should be connected at (or close to) the ground pin of the reference IC. The ferrite bead that connects the analog ground trace to system ground should also be placed close to the reference IC. [Figure 6-5](#) also shows the routing (from the top) of the dedicated analog input pair (V_P and V_N). These inputs are also routed as a differential pair.

The external reference IC should be placed as close as possible to the FPGA to reduce the opportunities for coupled noise and to minimize any impedances in the reference traces. The staggered via field also allows the 100 nF decoupling on V_{REFP} and V_{CCADC} to be placed in the center of the array close to the package balls. V_{REFP} should be decoupled to V_{REFN} and V_{CCADC} to GNDADC near the package balls.

When using the on-chip reference, the layout of the PCB is greatly simplified. The V_{REFP} and V_{REFN} pins should be shorted to AGND locally at the package balls. The ferrite beads used to separate AGND and digital GND should be placed close to the XADC balls in the center of the array along with a 100 nF decoupling capacitor for V_{CCADC} .

These suggestions are only intended to guide a low-cost PC board implementation. If it is feasible to create an analog reference plane, then there is no issue with doing this and it helps improve the ADC performance. However, the reference inputs should still be routed as differential pairs as shown. See the schematics and layout for the KC705 and AC701 evaluation boards for an example of a practical layout implementation. For more information, see XAPP554, *XADC Layout Guidelines Application Note* [\[Ref 6\]](#).

This section explains in more detail the example shown in [Example Instantiation, page 15](#). The Verilog and VHDL source code and test benches for this example can be found in the ZIP file [ug480_7Series_XADC.zip](#) that accompanies this user guide:

[Design file for UG480 ug480_7Series_XADC.zip](#)

In addition to instantiating a basic design, the use of the Analog Stimulus file as a means to introduce analog signals into the simulation is also explained. Finally, the output of a simple behavioral simulation of the design is shown. The project illustrates an instantiation and simulation of the XADC in Vivado.

One of the easiest methods to add the XADC to a design is to use the XADC instantiation wizard in the CORE Generator™ tool. In Vivado tools, the XADC Wizard can be found under the IP Catalog. The wizard is a GUI that allows designers to select the required block I/O and initialize the control registers for the required operation.

For more information on the XADC wizard, see PG091, *XADC Wizard LogiCORE IP Product Guide for Vivado Design Suite* [\[Ref 5\]](#).

The following HDL example sets up the XADC to monitor all the FPGA on-chip sensors, that is, Temperature, V_{CCINT} , V_{CCAUX} , and V_{CCBRAM} of a 7 series FPGA. See [Temperature Sensor, page 17](#) and [Power Supply Sensors, page 17](#). In addition, four auxiliary analog input channels are also monitored. The XADC is also set to automatically generate alarm outputs when the defined operating ranges for the FPGA supply voltages and temperature are exceeded (see [Automatic Alarms, page 56](#)). The XADC is operated in continuous sequence mode for this example (see [Sequencer Modes, page 49](#)). For clarity (and shorter simulations), the averaging function is disabled by the test bench. Averaging does not have any impact on the simulation results because an ideal model of the XADC is used. It is recommended that averaging is enabled when monitoring the on-chip sensors in a typical application to minimize any noise impacts. This is especially true if the automatic alarm functions are used. Averaging has been enabled for the on-chip sensors in the instantiation shown below.

During simulation, if a design uses calibration, calibration coefficients will assume ideal conditions.

Apart from initializing the alarm threshold registers and the automatic channel sequencer register, the configuration registers need to be initialized to enable alarm outputs, sequencer modes, and the ADC clock divider (see [Configuration Registers \(40h to](#)

[42h](#)), [page 36](#) for more information). Here is an instantiation in Verilog of the XADC example design:

```
`timescale 1ns / 1ps
module ug480 (
input DCLK, // Clock input for DRP
input RESET,
input [3:0] VAUXP, VAUXN, // Auxiliary analog channel inputs
input VP, VN, // Dedicated and Hardwired Analog Input Pair
output reg [15:0] MEASURED_TEMP, MEASURED_VCCINT,
output reg [15:0] MEASURED_VCCAUX, MEASURED_VCCBRAM,
output reg [15:0] MEASURED_AUX0, MEASURED_AUX1,
output reg [15:0] MEASURED_AUX2, MEASURED_AUX3,
output wire [7:0] ALM,
output wire [4:0] CHANNEL,
output wire OT,
output wire EOC,
output wire EOS
);
wire busy;
wire [5:0] channel;
wire drdy;
reg [6:0] daddr;
reg [15:0] di_drp;
wire [15:0] do_drp;
wire [15:0] vauxp_active;
wire [15:0] vauxn_active;
wire eos;
wire eoc;
reg [1:0] den_reg;
reg [1:0] dwe_reg;
reg [7:0] state = init_read;
parameter init_read = 8'h00,
read_waitdrdy = 8'h01,
write_waitdrdy = 8'h03,
read_reg00 = 8'h04,
reg00_waitdrdy = 8'h05,
read_reg01 = 8'h06,
reg01_waitdrdy = 8'h07,
read_reg02 = 8'h08,
reg02_waitdrdy = 8'h09,
read_reg06 = 8'h0a,
reg06_waitdrdy = 8'h0b,
read_reg10 = 8'h0c,
reg10_waitdrdy = 8'h0d,
read_reg11 = 8'h0e,
reg11_waitdrdy = 8'h0f,
read_reg12 = 8'h10,
reg12_waitdrdy = 8'h11,
read_reg13 = 8'h12,
reg13_waitdrdy = 8'h13;
always @(posedge DCLK)
if (RESET) begin
state <= init_read;
```



```
den_reg <= 2'h0;
dwe_reg <= 2'h0;
di_drp <= 16'h0000;
end
else
case (state)
init_read : begin
daddr = 7'h40;
den_reg = 2'h2; // performing read
if (busy == 0 ) state <= read_waitdrdy;
end
read_waitdrdy :
if (drdy ==1) begin
di_drp = do_drp & 16'h03_FF; //Clearing AVG bits for Configreg0
daddr = 7'h40;
den_reg = 2'h2;
dwe_reg = 2'h2; // performing write
state = write_waitdrdy;
end
else begin
den_reg = { 1'b0, den_reg[1] } ;
dwe_reg = { 1'b0, dwe_reg[1] } ;
state = state;
end
write_waitdrdy :
if (drdy ==1) begin
state = read_reg00;
end
else begin
den_reg = { 1'b0, den_reg[1] } ;
dwe_reg = { 1'b0, dwe_reg[1] } ;
state = state;
end
read_reg00 : begin
daddr = 7'h00;
den_reg = 2'h2; // performing read
if (eos == 1) state <=reg00_waitdrdy;
end
reg00_waitdrdy :
if (drdy ==1) begin
MEASURED_TEMP = do_drp;
state <=read_reg01;
end
else begin
den_reg = { 1'b0, den_reg[1] } ;
dwe_reg = { 1'b0, dwe_reg[1] } ;
state = state;
end
read_reg01 : begin
daddr = 7'h01;
den_reg = 2'h2; // performing read
state <=reg01_waitdrdy;
```

```
end
reg01_waitdrdy :
if (drdy ==1) begin
MEASURED_VCCINT = do_drp;
state <=read_reg02;
end
else begin
den_reg = { 1'b0, den_reg[1] } ;
dwe_reg = { 1'b0, dwe_reg[1] } ;
state = state;
end
read_reg02 : begin
daddr  = 7'h02;
den_reg = 2'h2; // performing read
state  <=reg02_waitdrdy;
end
reg02_waitdrdy :
if (drdy ==1) begin
MEASURED_VCCAUX = do_drp;
state <=read_reg06;
end
else begin
den_reg = { 1'b0, den_reg[1] } ;
dwe_reg = { 1'b0, dwe_reg[1] } ;
state = state;
end
read_reg06 : begin
daddr  = 7'h06;
den_reg = 2'h2; // performing read
state  <=reg06_waitdrdy;
end
reg06_waitdrdy :
if (drdy ==1) begin
MEASURED_VCCBRAM = do_drp;
state <= read_reg10;
end
else begin
den_reg = { 1'b0, den_reg[1] } ;
dwe_reg = { 1'b0, dwe_reg[1] } ;
state = state;
end
read_reg10 : begin
daddr  = 7'h10;
den_reg = 2'h2; // performing read
state  <= reg10_waitdrdy;
end
reg10_waitdrdy :
if (drdy ==1) begin
MEASURED_AUX0 = do_drp;
state <= read_reg11;
end
else begin
```

```
den_reg = { 1'b0, den_reg[1] } ;
dwe_reg = { 1'b0, dwe_reg[1] } ;
state = state;
end
read_reg11 : begin
daddr  = 7'h11;
den_reg = 2'h2; // performing read
state  <= reg11_waitdrdy;
end
reg11_waitdrdy :
if (drdy ==1) begin
MEASURED_AUX1 = do_drp;
state <= read_reg12;
end
else begin
den_reg = { 1'b0, den_reg[1] } ;
dwe_reg = { 1'b0, dwe_reg[1] } ;
state = state;
end
read_reg12 : begin
daddr  = 7'h12;
den_reg = 2'h2; // performing read
state  <= reg12_waitdrdy;
end
reg12_waitdrdy :
if (drdy ==1) begin
MEASURED_AUX2= do_drp;
state <= read_reg13;
end
else begin
den_reg = { 1'b0, den_reg[1] } ;
dwe_reg = { 1'b0, dwe_reg[1] } ;
state = state;
end
read_reg13 : begin
daddr  = 7'h13;
den_reg = 2'h2; // performing read
state  <= reg13_waitdrdy;
end
reg13_waitdrdy :
if (drdy ==1) begin
MEASURED_AUX3= do_drp;
state <=read_reg00;
daddr  = 7'h00;
end
else begin
den_reg = { 1'b0, den_reg[1] } ;
dwe_reg = { 1'b0, dwe_reg[1] } ;
state = state;
end
endcase
XADC #((// Initializing the XADC Control Registers
```

```

.INIT_40(16'h9000),// averaging of 16 selected for external channels
.INIT_41(16'h2ef0),// Continuous Seq Mode, Disable unused ALMs, Enable calibration
.INIT_42(16'h0400),// Set DCLK divides
.INIT_48(16'h4701),// CHSEL1 - enable Temp VCCINT, VCCAUX, VCCBRAM, and calibration
.INIT_49(16'h000f),// CHSEL2 - enable aux analog channels 0 - 3
.INIT_4A(16'h0000),// SEQAVG1 disabled
.INIT_4B(16'h0000),// SEQAVG2 disabled
.INIT_4C(16'h0000),// SEQINMODE0
.INIT_4D(16'h0000),// SEQINMODE1
.INIT_4E(16'h0000),// SEQACQ0
.INIT_4F(16'h0000),// SEQACQ1
.INIT_50(16'hb5ed),// Temp upper alarm trigger 85°C
.INIT_51(16'h5999),// Vccint upper alarm limit 1.05V
.INIT_52(16'hA147),// Vccaux upper alarm limit 1.89V
.INIT_53(16'h0000),// OT upper alarm limit 125°C using automatic shutdown - see Thermal Management
.INIT_54(16'ha93a),// Temp lower alarm reset 60°C
.INIT_55(16'h5111),// Vccint lower alarm limit 0.95V
.INIT_56(16'h91Eb),// Vccaux lower alarm limit 1.71V
.INIT_57(16'hae4e),// OT lower alarm reset 70°C - see Thermal Management
.INIT_58(16'h5999),// VCCBRAM upper alarm limit 1.05V
.SIM_MONITOR_FILE("design.txt")// Analog Stimulus file for simulation
)
XADC_INST (// Connect up instance IO. See UG480 for port descriptions
.CONVST (1'b0),// not used
.CONVSTCLK (1'b0), // not used
.DADDR (daddr),
.DCLK (DCLK),
.DEN (den_reg[0]),
.DI (di_drp),
.DWE (dwe_reg[0]),
.RESET (RESET),
.VAUXN (vauxn_active ),
.VAUXP (vauxp_active ),
.ALM (ALM),
.BUSY (busy),
.CHANNEL(CHANNEL),
.DO (do_drp),
.DRDY (drdy),
.EOC (EOC),
.EOS (EOS),
.JTAGBUSY (),// not used
.JTAGLOCKED (),// not used
.JTAGMODIFIED (),// not used
.OT (OT),
.MUXADDR (),// not used
.VP (VP),
.VN (VN)
);
assign vauxp_active = {12'h000, VAUXP[3:0]};
assign vauxn_active = {12'h000, VAUXN[3:0]};
assign EOC = eoc;
assign EOS = eos;

```

endmodule

The following is a simple test bench that sets up a DCLK of 50 MHz and then reads the status registers (conversion results) at the end of a sequence when EOS goes High. The test bench also does a write to the DRP after the reset to disable the averaging on the sensor channels. The test bench monitors the alarm outputs so alarm events can be seen. Analog signals are not introduced into the simulation using the test bench. Analog signals are read from a file by the simulation model. The SIM_MONITOR_FILE attribute used in the XADC instantiation points the model to the location of this file known as the *Analog Stimulus file*. The Analog Stimulus file used for this example is shown here:

TIME	VAUXP[0]	VAUXN[0]	VAUXP[1]	VAUXN[1]	VAUXP[2]	VAUXN[2]	VAUXP[3]	VAUXN[3]	Temp	VCCINT	VCCAUX	VCCBRAM
00000	0.005	0.0	0.200	0.0	0.50	0.0	0.1	0.0	25	1.00	1.80	1.00
34000	0.020	0.0	0.400	0.0	0.49	0.0	0.2	0.0	85	1.05	1.90	1.05
67000	0.049	0.0	0.600	0.0	0.51	0.0	0.5	0.0	20	0.95	1.71	0.95
99999	0.034	0.0	0.900	0.0	0.53	0.0	0.0	0.0	125	1.00	1.80	1.00

The format of the analog stimulus file is based on space- or tab-delimited data and can be created in a spreadsheet, for example. Many tools such as SPICE simulators or even equipment such as oscilloscopes export comma-separated value (CSV) formats, which are manipulated in a spreadsheet to generate an analog stimulus file for simulation. All time stamp information must be listed in the first column. Other columns list the analog values for the on-chip sensors and external analog inputs. The order of the columns is not important, the only requirement is that time stamp information is listed in the first column. For every time stamp added to the first column, a corresponding value is added to the other columns. Only the required analog input channel columns need to be listed. In this example, only the on-chip sensors and auxiliary channels 0 to 3 are listed in the analog stimulus file. In this stimulus file, the voltage on V_{CCAUX} moves from nominal supply (1.8V) to 1.9V at 27 μ s after the start of the simulation. The V_{CCAUX_ALARM} output can be seen to go active-High shortly after this event when the V_{CCAUX} supply is measured by the ADC during the sequence (see [Figure 6-6, page 77](#)). The upper alarm threshold for V_{CCAUX} has been set at 1.89V in the instantiation above.

```
`timescale 1ns / 1ps
module ug480_tb;
reg [3:0]  VAUXP, VAUXN;
reg      RESET;
reg      DCLK;
wire [15:0] MEASURED_TEMP, MEASURED_VCCINT, MEASURED_VCCAUX;
wire [15:0] MEASURED_VCCBRAM, MEASURED_AUX0, MEASURED_AUX1;
wire [15:0] MEASURED_AUX2, MEASURED_AUX3;
wire [15:0] ALM;
initial
begin
DCLK = 0;
RESET = 1;
#100 RESET = 0;
end
always #(10) DCLK= ~DCLK;
// Instantiate the Unit Under Test (UUT)
ug480 uut (
.VAUXP (VAUXP),
.VAUXN (VAUXN),
.RESET (RESET),
.ALM (ALM),
.DCLK (DCLK),
.MEASURED_TEMP (MEASURED_TEMP),
.MEASURED_VCCINT (MEASURED_VCCINT),
.MEASURED_VCCAUX (MEASURED_VCCAUX),
.MEASURED_VCCBRAM (MEASURED_VCCBRAM),
.MEASURED_AUX0 (MEASURED_AUX0),
```

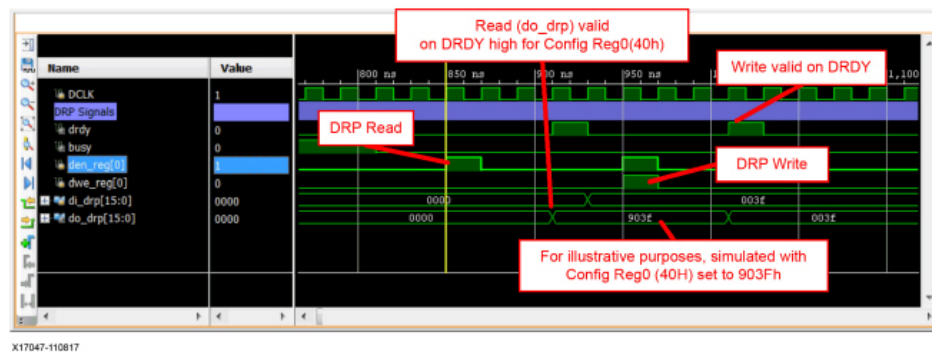
```
.MEASURED_AUX1 (MEASURED_AUX1),
.MEASURED_AUX2 (MEASURED_AUX2),
.MEASURED_AUX3 (MEASURED_AUX3)
);
endmodule
```

The simulation output shown in [Figure 6-6](#) shows two passes through the user-defined sequence in continuous sampling mode. The channels monitored in the sequence can be seen by looking at the CHANNEL[4:0] bus. The sequence is 8, 0, 1, 2, 3, 6, 10, 11, 12, and 13, which corresponds to Calibration, Temperature, V_{CCINT} , V_{CCAUX} , V_{CCBRAM} , Aux0, Aux1, Aux2, and Aux3. Then the sequence repeats. Note how the calibration channel takes longer to complete than the other channels. This is because the calibration routine involves three conversions (measurements) using the ADCs.



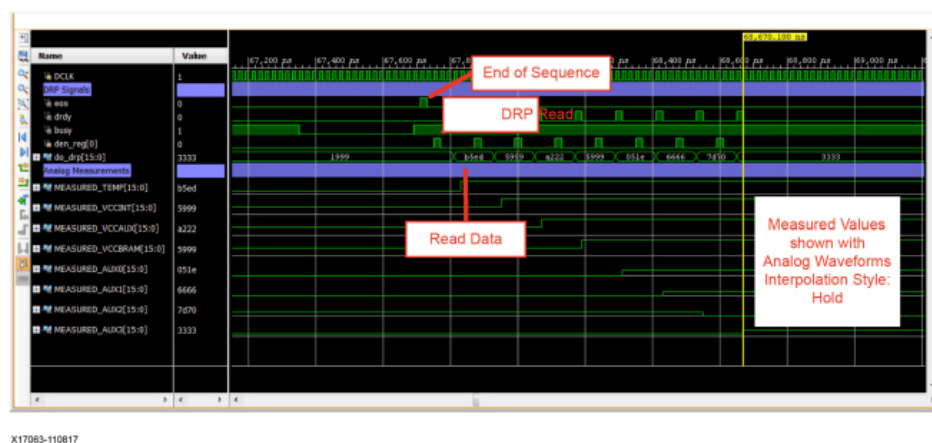
XADC Simulation Output (iSim)

After RESET goes Low, the test bench carries out a DRP write to register 40h (Configuration Register 0) to set the AVG1 and AVG0 bits to 00 (see [Figure 6-7](#)). This disables the averaging functionality for simulation.



DRP Write

When the EOS signal pulses high for one DCLK period at the end of a sequence (at approximately 30 μ s) the test bench reads the Status registers (see [Figure 6-8](#)). Note that the simulation model uses the full 16 ADC conversion result because it is an ideal model of the ADC. Thus for example, the result for the V_{CCINT} measurement is 5555h, which corresponds to 1V. In [Figure 2-10, page 27 \(Ideal Power Supply Transfer Function\)](#) 1V is shown as 555h. This is a 12-bit MSB justified result. However, the 4 LSBs of the Status register also contain data that would be 5h if the ADC was an ideal 16-bit ADC.



DRP Read of Status Register at EOS

Note how the V_{CCAUX_ALARM} output goes high during the second pass through the sequence ([Figure 6-6](#)). The alarm goes active at the end of the conversion on the V_{CCAUX} channel when the result is loaded in the Status register. The result is read from the Status register by the test bench when EOS next goes high. The V_{CCAUX} is ~1.9V (as set in the stimulus file) and is greater than the 1.89V limit set when the XADC was instantiated in the design which activates the alarm.

Additional Resources and Legal Notices

Appendix 7

Additional Resources and Legal Notices

Xilinx Resources

For support resources such as Answers, Documentation, Downloads, and Forums, see [Xilinx Support](#) .

Solution Centers

See the [Xilinx Solution Centers](#) for support on devices, software tools, and intellectual property at all stages of the design cycle. Topics include design assistance, advisories, and troubleshooting tips.

Documentation Navigator and Design Hubs

Xilinx[®] Documentation Navigator provides access to Xilinx documents, videos, and support resources, which you can filter and search to find information. To open the Xilinx Documentation Navigator (DocNav):

- From the Vivado[®] IDE, select **Help > Documentation and Tutorials** .
- On Windows, select **Start > All Programs > Xilinx Design Tools > DocNav** .
- At the Linux command prompt, enter `docnav` .

Xilinx Design Hubs provide links to documentation organized by design tasks and other topics, which you can use to learn key concepts and address frequently asked questions. To access the Design Hubs:

- In the Xilinx Documentation Navigator, click the **Design Hubs View** tab.
- On the Xilinx website, see the [Design Hubs](#) page.

Note: For more information on Documentation Navigator, see the [Documentation Navigator](#) page on the Xilinx website.

References

Additional material useful to this document can be found here:

1. ([PG019](#)), *LogiCORE IP AXI XADC (v1.0) Product Guide*
2. ([UG475](#)), *7 Series FPGAs Packaging and Pinout Product Specifications User Guide*
3. ([UG470](#)), *7 Series FPGAs Configuration User Guide*
4. ([UG585](#)), *Zynq-7000 All Programmable SoC Technical Reference Manual*
5. ([PG091](#)), *XADC Wizard LogiCORE IP Product Guide for Vivado Design Suite*
6. ([XAPP554](#)), *XADC Layout Guidelines Application Note*
7. ([XAPP795](#)), *Driving the Xilinx Analog-to-Digital Converter Application Note*
8. ([UG908](#)), *Vivado Design Suite User Guide: Programming and Debugging*
9. ([WP442](#)), *Efficient Implementation of Analog Signal Processing Functions in Xilinx Devices*
10. ([UG865](#)), *Zynq-7000 SoC Packaging and Pinout Product Specification User Guide*
11. ([DS180](#)), *7 Series FPGAs Overview*

Revision History

The following table shows the revision history for this document.

Date	Version	Revision
03/01/2011	1.0	Initial Xilinx release.
03/28/2011	1.1	Added “Dual 12-Bit MSPS Analog-to-Digital Converter” to document title. Modified first paragraph and added second paragraph in Chapter 1, Introduction and Quick Start . Added Table 1-1 . Removed the <i>Thermal Diode (DXP and DXN)</i> section from Chapter 4.
10/25/2012	1.2	This version was updated to include information for the Zynq-7000 SoC devices: Added “Zynq-7000 SoC” to document title. In Figure 1-1 Zynq-7000 SoC information was added and the control and status registers changed from 32 x 16 bits to 64 x 16 bits. The <i>System Monitor Support</i> section changed to Differences between Virtex-5 and Virtex-6 System Monitors . Functionality for the XADC block in 7 series FPGAs is now defined for previously undefined status registers in subsequent chapters. Capacitor values in Figure 1-2 changed, and a note was added about the placement of 100 nF decoupling capacitors. In Table 1-1 , for V_{REFP_0} , GND changed to GNDADC, accurate reference <i>source</i> changed to accurate reference <i>IC</i> , and V_{P_0} and V

N₀ types were changed to *dedicated* analog inputs. Footnotes were added to support 7 series and Zynq device pin packages. A note about application guidelines was added after [Table 1-1](#). Information about auxiliary analog channels was added to the [External Analog Inputs](#) section. In [Table 1-2](#), ports ALM[0] through [ALM[3] have XADC removed from the description. Alarms specific to Zynq-7000 SoC devices (ALM[4], ALM[5], and ALM[6]) were added. The example in [Example Instantiation](#) was updated. In [ADC and Sensors](#), information about an analog input signal of 200 mV was modified, and [Equation 1-1](#) was added. A new [Zynq-7000 SoC](#) subsection was added to the end of the chapter.

In [ADC Transfer Functions](#) MSBs are defined as left-most bits. In [Auxiliary Analog Inputs in the instantiation](#) was changed to *on the primitive*, and information that configuration is automatic when analog inputs are connected was added. The Note on [page 22](#) was expanded to clarify device support of auxiliary analog channels. [Equation 2-2](#) was added (unipolar mode), [Equation 2-1](#), [Equation 2-3](#), and [Equation 2-4](#) were modified (“10” was changed to “9”), and the paragraph following [Equation 2-4](#) was added. Information about additional external resistance was added to [Unipolar Input Signals](#). In [Temperature Sensor](#) this sentence was deleted: “The on-chip temperature sensor has a maximum measurement error of $\pm 4^{\circ}\text{C}$ over a range of -40°C to $+125^{\circ}\text{C}$.” Information was added to the [Power Supply Sensor](#) section about which supplies are monitored and where measurements are stored for Zynq-7000 SoC devices.

10/25/2012

1.2
(Cont'd)

Min/max register lists were updated for Zynq-7000 SoC devices in [Figure 3-1](#). In [Table 3-1](#) the V_{REFN} description was updated, new Zynq-7000 device channels V_{CCPINT}, V_{CCPAUX}, and V_{CCO_DDR} were added. Zynq-7000 SoC

status registers were added to [Figure 3-1](#) . Flag registers for DI5 through DI8 in [Figure 3-2](#) changed and DIS was removed from [Table 3-2](#) . [XADC Calibration Coefficients](#) were added. Configuration registers ALM[4], ALM[5], and ALM[6] were added to [Figure 3-4](#) . In [Table 3-3](#) , a new row was added for bits DI19 to DI11. In [Table 3-7](#) , rows for ADC channels 13, 14, and 15 were added. Zynq-7000 SoC alarm threshold register information was added to [Alarm Registers \(50h to 5Fh \)](#) . A note in the section [DRP JTAG Interface](#) mentions conditions in which the external JTAG access is disabled for Zynq devices. Section [Zynq-7000 SoC Processing System \(PS\) to XADC Dedicated Interface](#) was added to the end of the chapter.

In [XADC Operating Modes](#) , *operate both ADCs in parallel* in the first paragraph changed to *operate both ADCs in lock step* . In [Single Channel Mode](#) , the first sentence was deleted (“Single channel mode is the most basic way users can modify the operation of the XADC.”) Sequence numbers and descriptions in [Table 4-1](#) and [Table 4-2](#) changed and were added for Zynq-7000 SoC devices. In [Sequencer Modes](#) , default mode sequences were changed in [Table 4-3](#) and added for Zynq-7000 SoC devices. The sentence before [Table 4-4](#) was deleted. The footnote in [Table 4-4](#) and [Table 4-6](#) was enhanced to mention auxiliary analog channel support. [Table 4-7](#) was updated to include Zynq-7000 SoC channels. Alarm threshold registers for the Zynq-7000 SoC were added to [Table 4-8](#) . A new section [XADC Enhanced Linearity Mode](#) and [Figure 4-5](#) were added to the end of the chapter.

The [Dynamic Reconfiguration Port \(DRP\) Timing](#) section was added. Removed [Table 5-1: XADC Timing Information](#).

In [Reference Inputs \(\$V_{REFP}\$ and \$V_{REFN}\$ \)](#) , added “Noise on the reference voltage also adds noise to the ADC conversion and results in more code

		transition noise or poorer than expected SNR” to the end of the first paragraph. Capacitor values were changed, a new 10 μF capacitor was added, and notes were added to Figure 6-1 . Notes were added to Figure 6-5 . XADC Software Support was completely replaced. In that section, the Verilog instantiation was replaced. Figure 6-6: Analog Stimulus File and Figure 6-7: XADC Simulation Output were deleted. In the new section, Figure 6-5 through Figure 6-8 were added.
03/10/2014	1.3	Updated the disclaimer and copyright. Changes to Figure 1-2 in XADC Pinout Requirements . In External Analog Inputs , clarified how auxiliary analog inputs are handled in Vivado tools. Improved description of RESET signal in Table 1-2 . In Adjusting the Acquisition Settling Time , added to description of ACQ bit. In Analog Input Description , added note to Equation 2-1 . Added Table 3-3 , Calibration Coefficients in Status Register. Added to description of DI4 and DI5 in Table 3-5 . Added to important note in DRP JTAG Interface. Added clarification to Table 4-1 and Table 4-2 in ADC Channel Selection Registers (48h and 49h) section. Modified Figure 4-3 . Added note to Table 4-8 , Alarm Threshold Registers. Changes to description of Thermal Management . Modified Figure 5-1 . Updated Conversion Phase and Event-Driven Sampling . Added to description of Acquisition Phase . Updated External Analog Inputs . Modified Figure 6-3 . Modified Example Design Instantiation and Example Design Test Bench . Updated Using XADC Instantiation Wizard . Updated Figure 6-7 .
04/9/2014	1.3.1	Updated link to UG585 in References .
05/31/2014	1.4	Updated Figure 1-2 , Figure 2-10 , and Figure 3-3 . Updated V_{REFP_0} package pin description in Table 1-1 . Updated Gain Coefficients and Single Pass Mode . Added second paragraph to